

Exact test problems and a work–accuracy protocol for benchmarking nearest correlation matrix solvers

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Abstract

Nearest correlation matrix (NCM) solvers are usually compared on matrices whose solutions are unknown, so the accuracy each delivers goes unmeasured. We specialize the known technique of generating test problems from prescribed optimality conditions to the NCM problem, obtaining instances with closed-form primal and dual solutions and independently controlled rank, multiplicity and eigenvalue separation. Measuring forward error against them changes what a comparison reports. The semismooth Newton method with the Borsdorf–Higham line search is cheapest under all 11 conventions examined, at a cost flat in dimension: five eigendecompositions at $n = 100$ and again at $n = 2105$. Second at $n = 100$, under every convention, is Anderson-accelerated alternating projections, absent from recent NCM comparisons. The ordering of the remaining methods is a property of the test set, changing with the accuracy target and above all with the solution rank: two published methods exchange places between $r = 20$ and $r = 50$ at $n = 100$, and that rank rises with dimension. Under the dimension-scaled rule $\|\nabla\theta\|_2 \leq 10^{-7}n$, first-order methods return about ten times less accuracy at $n = 500$ than at $n = 100$, so a larger problem can appear easier. A known finite-precision failure of objective-value Armijo tests recurs in a second published method, where a safeguard from the NCM literature removes it.

Keywords. nearest correlation matrix; semidefinite programming; test problem generation; benchmarking; semismooth Newton method; reproducibility.

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1 Introduction

Given a symmetric matrix $G \in \mathbb{R}^{n \times n}$, the nearest correlation matrix problem asks for

$$\min_X \frac{1}{2} \|X - G\|_F^2 \quad \text{subject to} \quad X \succeq 0, \text{diag}(X) = e. \quad (1)$$

The problem has a unique solution. Since Higham’s alternating-projections method [16], the literature has produced dual formulations and first-order schemes [23, 8], a quadratically convergent semismooth Newton method [26], its preconditioned and globalized refinement [6], accelerated projection methods [18], and, most recently, accelerated gradient methods with quasi-Newton preconditioning [19]. Most of these papers include numerical comparisons, and those comparisons are how a reader decides which method to use.

This paper is about how such comparisons are made. We ask a narrow question: *when NCM solvers are compared, which methodological choices change the conclusion, and which only change the numbers?* We identify four candidate confounds:

1. *Native stopping rules.* Each method stops when its own criterion is met, so the solvers stop at different accuracies.
2. *Approximate references.* Forward error is measured against the output of another solver instead of against the true solution.
3. *Rejected-iterate accounting.* Line-search trials that are rejected can be credited with reaching a target that the method has not actually reached.
4. *Hidden primitive work.* ‘Iterations’ count different operations in different methods, and some per-iteration work is not recorded at all.

A careful benchmark must control each of these. Without an exact solution, however, the second confound cannot be removed, and without trajectories the first and third cannot be separated.

Contributions. In brief, we give an exact-solution NCM family and a matched-accuracy protocol, and use them to show that conclusions drawn from NCM solver comparisons do not transfer across benchmark regimes: the rank at which an ordering established at $n = 100$ reverses moves higher at $n = 500$, the order also changes with the degree of invalidity on matrices built from equity returns, and four matrices from the literature are consistent with this dependence. We also show that a known finite-precision line-search defect recurs in a second published method and occurs on real matrices. Each contribution builds on established practice (section 1.1).

1. *An exact-solution NCM family* (section 2). We specialize the standard construction of test problems from prescribed optimality conditions [21, 9] to the NCM problem. For semidefinite programs, the same idea is due to Wei and Wolkowicz [28]. The parts specific to NCM are the projection identity $\Pi_{\mathcal{S}_+^n}(G + \text{Diag } y^*) = X^*$, independent control of rank, strict-complementarity defect and eigenvalue separation, a paired-seed design, sufficient conditions for $|G_{ij}| \leq 1$ together with a fixed-rank saturation result for $\max_{i \neq j} |X_{ij}^*|$, and released instance sets at $n = 100$ and $n = 500$.
2. *A matched-accuracy protocol for NCM* (section 3). We adapt data profiles [25], fixed-target evaluation [15] and work-precision diagrams [14] to spectral solvers. The unit of cost is one EVD, which we check against wall-clock time. Because the reference is exact, two rules become enforceable: rejected trials receive no accuracy credit, and accuracy is matched in forward error. Primitive work is reported separately.
3. *An invariant first place above a regime-dependent ordering* (section 5). Six verified variants of five methods were run on 270 instances at $n = 100$ and 270 at $n = 500$, with 135 further instances at higher rank, 125 built from two equity panels at $n = 550$ and $n = 2105$, and four matrices from the literature. The cheapest method is the same under all 11 conventions examined, and its cost is flat in dimension: 5 eigendecompositions at $n = 100$ and at $n = 2105$. Second at $n = 100$ is Anderson-accelerated alternating projections [18], absent from recent NCM comparisons; it keeps that place on the exact-solution families but not on perturbed or literature matrices. Below the first place the ordering changes with the accuracy target, with the solution rank, and with how far the input is from a correlation matrix. The rank dependence holds between published methods alone: Dykstra-APM and AGD-SDAJ-BH exchange

places between $r = 20$ and $r = 50$ at $n = 100$, the exchange has not occurred by $r = 150$ at $n = 500$, and at $n = 2105$ Anderson-APM and AGD-SDAJ-BH exchange places between $r = 200$ and $r = 400$. The same pattern appears between SBB-Dual and AGD-SDAJ-BH, whose exchange is bracketed between $r = 5$ and $r = 20$ at $n = 100$ and between $r = 50$ and $r = 75$ at $n = 500$. Only a family that varies one spectral quantity at a time makes such an association directly observable.

4. *Replication of a known line-search defect* (section 4). The failure of objective-value sufficient-decrease tests near a minimizer is well known [13]. For the NCM dual, Borsdorf analysed it [5, §4.7.1], and this led to the modified step selection of Borsdorf and Higham [6]. The new content is limited to two points. First, the defect recurs in a separately published first-order method [19]. Second, we quantify it on the controlled family: 56/270 stalls for semismooth Newton and 18/270 for AGD-SDAJ, both removed by the Borsdorf-Higham correction. We also document a second, simpler trap (a redundant Jacobian factorization).
5. *An NCM quantification of the stopping-rule confound* (section 6). Dolan, Moré and Munson [11] warned that native stopping tests confound comparisons. For NCM, the widely used rule $\|\nabla\theta\|_2 \leq 10^{-7}n$ leaves the achieved forward error spread over a factor of 383 across solvers at $n = 100$, and it degrades accuracy with dimension unequally across methods.

What we do not claim. We do not claim that any method is best in general. The ranking results are for a synthetic family at $n = 100$ and $n = 500$, with instances built from equity returns at $n = 550$ and $n = 2105$ (sections 5.8 and 5.9) and four matrices from the literature (section 5.10) as external checks, all on one machine. SBB-Dual is a new method from the author’s companion manuscript in preparation [20]. It is specified in full in section E, receives the same exposition as the others, and its convergence theory is not used here. No conclusion of this paper depends on it: without SBB-Dual, Newton-SIN-BH is still first, Anderson-APM is still second at $n = 100$, and the rank dependence is still present, between Dykstra-APM and AGD-SDAJ-BH.

1.1 Related work

Benchmarking methodology. Performance profiles [10] compare solvers by the ratio of their cost to the best cost on each problem. Cost is measured at each solver’s own termination, so the solver’s stopping test implicitly sets the accuracy. Dolan, Moré and Munson [11] showed that differing stopping criteria bias such profiles, and proposed rerunning under a common optimality measure. Data profiles [25] count cost in a problem-independent unit and show how rankings change with the required accuracy. COCO [15] uses a ladder of fixed targets against known optima. Work-precision diagrams, standard in the comparison of ODE solvers [14], plot error against a reference solution versus work. Gould and Scott [12] show that performance-profile rankings below the leading solver can change when solvers are added or removed. Beiranvand, Hare and Lucet [4] survey best practice, including known-solution test problems and reproducibility. Our protocol follows these principles; our contribution is to make their exact-reference versions possible for the NCM problem. The second/third-place reversals of section 5 arise with a *fixed* solver set, from the accuracy target and the instance regime, so the mechanism differs from the one studied by Gould and Scott.

Test problems with known solutions. Generating test problems from prescribed optimality conditions is standard for unconstrained optimization [24] and for quadratic programming [21, 9].

Wei and Wolkowicz [28] generate semidefinite programs from a chosen optimal primal–dual pair with prescribed complementarity nullity, and show that nullity correlates with solver difficulty. Our family applies the same idea to the NCM problem. Their generator does not apply directly: it builds a linear-objective SDP and chooses the constraint operator as part of the construction, whereas NCM fixes the constraint operator as $\text{diag}(X) = e$ and leaves only G free. As far as we have found, no NCM study uses instances with a known exact solution. The test sets of [16, 26, 5, 6, 18, 2] are random or data-derived, as are the Anymatrix CORRINV matrices [17] used in [19], and in all of them the reference solutions are computed approximately.

Finite-precision line searches. An objective-value sufficient-decrease test cannot be evaluated accurately near a minimizer, because the difference of function values suffers cancellation. Hager and Zhang [13] introduced approximate Wolfe conditions for this reason. For the NCM dual, Borsdorf [5, §4.7.1] derived the stagnation threshold, $\|\nabla\theta\|$ of the order of the square root of the rounding error in θ (about 10^{-8}), noted that the problem is general to Armijo backtracking, and proposed the alternatives that became the modified step selection of [6]. Section 4 replicates this known phenomenon; it does not rediscover it.

1.2 Outline

Section 2 constructs the instance family. Section 3 states the protocol, section 4 describes the two implementation traps, section 5 reports the ranking study, and section 6 examines the dimension-scaled stopping rule. Section 7 covers reproducibility and section 8 discusses limitations. The appendices contain the proofs for the instance family, the verification records for our reimplementations of AGD-SDAJ and Anderson-APM, the artifact manifest, and pseudocode for every solver.

2 The KKT-controlled instance family

2.1 Dual formulation

Following [23, 26], the Lagrangian dual of (1) is the unconstrained problem

$$\min_{y \in \mathbb{R}^n} \theta(y) := \frac{1}{2} \|\Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y))\|_F^2 - e^\top y, \quad \nabla\theta(y) = \text{diag}(\Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y))) - e. \quad (2)$$

Here $\Pi_{\mathcal{S}_+^n}$ is the projection onto the positive semidefinite cone. The gradient is globally Lipschitz with constant 1. If y^* satisfies $\nabla\theta(y^*) = 0$, then $X^* = \Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y^*))$ is the unique solution of (1). Each evaluation of θ or $\nabla\theta$ requires one symmetric eigendecomposition (EVD) of $C(y) := G + \text{Diag}(y)$.

2.2 Construction and exactness

The idea is to choose the solution first and then build a G for which that solution satisfies the KKT conditions exactly. This is the standard technique of generating instances from prescribed optimality conditions [21, 9], and in particular Wei and Wolkowicz’s generator of semidefinite programs with prescribed complementarity nullity [28], specialized here to the NCM problem. What is specific to NCM is the exact projection identity of proposition 2.2, the independent control of the three spectral quantities in section 2.3, and the entrywise conditions of section 2.4. Their construction does not cover the present problem: it generates the data of a *linear*-objective SDP and is free to choose the constraint operator itself (their Algorithm 2.3, Steps 3–4), whereas the NCM problem fixes both the quadratic objective and the operator $\mathcal{A} = \text{diag}$ with $b = e$, leaving

G as the only free data. The shared element is the idea of prescribing an optimal primal–dual pair and controlling the dimension of its common nullspace, their complementarity nullity, which corresponds to $|\beta|$ here.

Definition 2.1 (KKT-controlled instance). Let $n > r \geq 1$. Draw $U \in \mathbb{R}^{n \times r}$ with independent standard-normal entries and rescale every row to unit Euclidean norm. Set $X^* = UU^\top$, so that $\text{diag}(X^*) = e$, $X^* \succeq 0$ and $\text{rank } X^* = r$ almost surely. Let $Q_0 \in \mathbb{R}^{n \times (n-r)}$ have orthonormal columns spanning $\ker X^*$. For $\mu \in \mathbb{R}_{\geq 0}^{n-r}$ define

$$W = Q_0 \text{Diag}(\mu) Q_0^\top, \quad G = X^* - W + \text{Diag}(\text{diag } W), \quad y^* = -\text{diag}(W). \quad (3)$$

Proposition 2.2 (Exactness). *For every $\mu \geq 0$, $\Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y^*)) = X^*$ and $\nabla \theta(y^*) = 0$. Hence X^* is the unique nearest correlation matrix to G .*

Proof. By construction, $G + \text{Diag}(y^*) = X^* - W$. Both X^* and W are symmetric positive semidefinite, and $X^*W = X^*Q_0 \text{Diag}(\mu) Q_0^\top = 0$ because the columns of Q_0 lie in $\ker X^*$. The ranges of X^* and W are therefore orthogonal. They need not be complementary: if some $\mu_j = 0$, then $\text{range } W$ is a proper subspace of $\ker X^*$. Consequently $X^* - W$ has the spectral decomposition obtained by combining those of X^* and $-W$ on orthogonal subspaces. Its positive part is X^* , and its nonpositive part is $-W$. Thus $\Pi_{\mathcal{S}_+^n}(X^* - W) = X^*$, and $\nabla \theta(y^*) = \text{diag}(X^*) - e = 0$. Since θ is convex and differentiable, y^* minimizes (2). Uniqueness of the primal solution is classical [16, 26]. $\square$

In floating point, the generator recomputes the projection rather than relying on the identity. Over the 270 released instances, the KKT residual is about 1.6×10^{-14} and the exactness error $\|\Pi_{\mathcal{S}_+^n}(C(y^*)) - X^*\|_F$ is about 8.2×10^{-14} .

2.3 Prescribed spectral structure

Because the ranges are orthogonal, the spectrum of $C(y^*) = X^* - W$ is the multiset union

$$\text{spec } C(y^*) = \{\lambda_1(X^*), \dots, \lambda_r(X^*)\} \cup \{-\mu_1, \dots, -\mu_{n-r}\}. \quad (4)$$

In the notation of [26], let α, β, γ index the positive, zero and negative eigenvalues of $C(y^*)$. Then

- $|\alpha| = r$, so the rank of the solution is prescribed;
- $|\beta| = \#\{j : \mu_j = 0\}$ exactly, so the multiplicity of zero eigenvalues at the solution is prescribed;
- an entry $\mu_j = \delta > 0$ produces an eigenvalue $-\delta$, which moves toward the nonsmooth boundary of $\Pi_{\mathcal{S}_+^n}$ as $\delta \rightarrow 0$.

Terminology. Qi and Sun [26] showed that constraint nondegeneracy always holds for the NCM problem, so the family does not, and could not, vary constraint degeneracy. What it varies is *failure of strict complementarity*: zero eigenvalues of $C(y^*)$ ($|\beta| > 0$), and near-zero eigenvalues (small δ) that approach it. Throughout this paper, a *degeneracy cell* means one (r, m, δ) combination in this sense, and the *strict-complementarity defect* means $|\beta|$. The generator uses $\mu = (\mu_{\text{bulk}}, \dots, \mu_{\text{bulk}}, \delta, \dots, \delta)$ with m copies of δ and $\mu_{\text{bulk}} = 0.1$. The separation parameter is $\text{sep} = \delta/\mu_{\text{bulk}}$ (we avoid ρ , which denotes the line-search backtracking factor in sections 4 and 5). The released grid is

$$r \in \{5, 20, 50\}, \quad m \in \{1, 5, 20\}, \quad \delta \in \{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}, 0\},$$

with five paired seeds per rank. This gives $3 \times 3 \times 6 = 54$ cells and 270 instances at $n = 100$. In plots, sep is shown on a $\log_{10}$ axis and the case $\text{sep} = 0$ is shown separately.

2.4 Correlation-like inputs: sufficient conditions for $|G_{ij}| \leq 1$

Proposition 2.2 places no restriction on the entries of G . For the instances to resemble invalid correlation matrices, which is the setting in which NCM problems arise, the generator additionally requires $\text{diag}(G) = e$ (true by construction), $\lambda_{\min}(G) < 0$, and $|G_{ij}| \leq 1$. The first two are checked directly. The results below explain when the third can be expected to hold. Let P_r be the orthogonal projector onto $\text{range } U$, so that $Q_0 Q_0^\top = I - P_r$, and define

$$a := \max_{i \neq j} |X_{ij}^*|, \quad b := \max_{i \neq j} |(P_r)_{ij}|.$$

Proposition 2.3 (Deterministic bound, uniform μ). *If $\mu = c e$ with $c > 0$, then $|G_{ij}| \leq a + cb$ for all $i \neq j$. In particular, if $a < 1$, then $0 < c \leq (1 - a)/b$ is sufficient for $|G_{ij}| \leq 1$.*

Proposition 2.4 (Fixed-rank saturation). *Fix $r \geq 2$ and let the rows u_i of U be independent and uniform on S^{r-1} . Then $a \rightarrow 1$ as $n \rightarrow \infty$, along every realization.*

Proposition 2.4 says only that a saturates. It does *not* imply that a fixed $c > 0$ eventually violates proposition 2.3: at fixed r the quantity b also decreases with n , so whether $a + cb$ exceeds 1 depends on the relative rates of $1 - a$ and b , which we do not establish. We therefore make no asymptotic feasibility claim, and the generator relies on entrywise screening.

Proposition 2.5 (High-dimensional regime). *Let the rows of U be independent and uniform on S^{r-1} and let $\eta \in (0, 1)$ be a failure probability (distinct from the separation parameter δ of proposition 2.1). Then with probability at least $1 - \eta$,*

$$a \leq \sqrt{\frac{2}{r} \left(2 \log n + \log \frac{2}{\eta} \right)}.$$

In particular $a \rightarrow 0$ whenever $r / \log n \rightarrow \infty$.

The bound follows from the subgaussianity of the uniform distribution on the sphere [27, Thm. 3.4.5] and a union bound; see section A. We state it only for a , the quantity that governs proposition 2.3 in the regime of interest. A companion bound on b would require a model for $\text{range } U$ that we do not need here, since the released family lies in the fixed-rank regime of proposition 2.4, where proposition 2.5 does not apply.

Proofs are given in section A. These results give sufficient conditions only. Failure of the bound in proposition 2.3 does not show that some $|G_{ij}|$ exceeds 1, and for that reason the generator checks every entry directly instead of relying on the bound. Proposition 2.3 assumes uniform μ , whereas the released instances use a mixed μ (bulk value 0.1 together with m copies of δ). The extension to mixed μ is left open: the extra off-diagonal term it introduces depends on the choice of orthonormal basis for $\ker X^*$ and is not determined by (n, r, m, δ) alone (section A). The released mixed- μ instances are therefore screened entrywise rather than certified by a bound. None of these results says anything about indefiniteness of G , which is checked separately.

Which regime the released family is in. Table 1 reports empirical values of (a, b) at $n = 100$ over 30 seeds per rank. The family is in the regime of proposition 2.4, not proposition 2.5. The rank is fixed and small relative to n , a is close to 1 at $r = 5$, and the bound on c implied by proposition 2.3 shrinks sharply as the rank decreases.

The table is consistent with the screening outcomes. At $\mu_{\text{bulk}} = 0.1$, the mean sufficient bound at $r = 5$ (0.162) exceeds 0.1, but the worst seeds ($\max a = 0.9981$) fall below it. In the paired

Table 1: Empirical (a, b) at $n = 100$, 30 seeds per rank. The last column is the mean over seeds of the sufficient bound $(1 - a)/b$ from proposition 2.3.

r	mean a	max a	mean b	max b	mean $(1 - a)/b$
5	0.9905	0.9981	0.0590	0.0708	0.162
20	0.7495	0.8340	0.1483	0.1871	1.690
50	0.5081	0.5719	0.1856	0.2083	2.650

design (section 2.5), 5 of the 7 candidate seeds were accepted at $r = 5$; the 2 rejections were for $|G_{ij}| > 1$. At $r = 20$ and $r = 50$, 5 of 5 were accepted. With $\mu_{\text{bulk}} = 1$, the sufficient bound fails at every rank tested, and in an earlier design every $r = 5$ candidate was rejected. This is why the generator uses $\mu_{\text{bulk}} = 0.1$.

2.5 Paired-seed design and screening

Each candidate seed is screened once per rank against its entire (m, δ) block. A seed is accepted only if all 3×6 instances built from it pass the checks ($\text{diag}(G) = e$, $|G_{ij}| \leq 1$, $\lambda_{\min}(G) < 0$, and a recomputed KKT residual and exactness error each below 10^{-9}). The accepted seeds are then reused across every (m, δ) cell. Varying the separation therefore changes only μ , with U held fixed, so comparisons along δ are within-seed. Rejections are counted and reported, not silently discarded.

Because of the screening, the $r = 5$ sample is conditioned differently from the $r = 20$ and $r = 50$ samples: at $r = 5$ it excludes the seeds with the largest a . Comparisons across ranks should be read with this in mind.

2.6 Release

We release the 270 paired instances (G , X^* , y^* , SHA-256 hashes of the upper triangle of G , and per-instance screening diagnostics), the generator, and the seed-screening log. Section D lists the files. We also release a paired set at $n = 500$, generated on the same grid. All 270 instances passed screening; seed acceptance was 5 of 7 candidates at $r = 5$ (both rejections for $|G_{ij}| > 1$) and 5 of 5 at $r = 20$ and $r = 50$, with $\max_{i \neq j} |G_{ij}| = 0.99987$ among the accepted $r = 5$ instances. The ranking study of section 5.6 uses all five paired seeds per rank.

3 Protocol

This section sets out the measurement protocol. Each rule addresses a confound that we observed on this family, and each is stated together with the evidence that motivates it.

3.1 Native stopping rules as a confound

Suppose every solver is run to its own exit criterion and the resulting costs are compared. The comparison then credits a method for stopping early as much as for converging quickly. This happens even when all methods share one nominal rule. Under the dimension-scaled rule $\|\nabla\theta\|_2 \leq 10^{-7}n$ used by [19], which at $n = 100$ is 10^{-5} , the six solvers of section 5 and the Newton-SIN control stop at median forward errors between 5.04×10^{-8} and 1.93×10^{-5} , a factor of 383 (section 6, fig. 5). A cost comparison taken at these exits compares different accuracies.

Table 2: Median primitive work at matched tolerance 10^{-11} , 270 instances, $n = 100$. An ‘outer iteration’ denotes a different amount of work in each row.

solver	EVDs	CG iterations	line-search trials	accepted outer its
Newton-SIN-BH	5	24	0	4
Newton-SIN	11	29.5	0	5
AGD-SDAJ-BH	70	0	14	9
AGD-SDAJ	81.5	0	25	10
SBB-Dual	46.5	0	0	45.5
Dykstra-APM	102.5	0	0	102.5
Anderson-APM	34	0	0	34

3.2 Work–accuracy trajectories against the exact solution

We take one EVD as the unit of cost. After *every* EVD, whether it produces an iterate, a line-search trial or a Jacobian factorization, we record the pair

$$(\text{cumulative EVDs}, \|X_k - X^*\|_F),$$

together with $\|\nabla\theta\|_2$ and a flag indicating whether the evaluated point was accepted. Here X^* is the exact solution from proposition 2.2, not the output of another solver. All runs use a common, tight tolerance of 10^{-11} and a budget of 4000 EVDs. These trajectories are work–precision diagrams in the sense of [14], and reading costs off them at a ladder of targets follows data profiles [25] and fixed-target evaluation [15]. The difference is that the reference is the exact solution, not the best value found by any solver. For any target ε , the cost can then be read off the trajectory after the run. We define the *cost to ε* as the cumulative EVD count at the first accepted iterate that satisfies $\|X_k - X^*\|_F \leq \varepsilon$ and continues to satisfy it for every later accepted iterate. A solver that never reaches the target, or reaches it and later loses it, is recorded as not reaching it at that ε .

Trajectories also expose *plateaus*. The cost of a quadratically convergent method takes very few distinct values across many decades of ε : in table 5, Newton-SIN-BH costs 3, 4 or 5 EVDs at every target from 10^{-2} to 10^{-10} . A single-tolerance comparison therefore depends on where the tolerance falls relative to these plateaus.

3.3 Accepted-only accounting

A rejected line-search trial always costs one EVD. Under *accepted-only* accounting it can never be credited with reaching a target, because the method did not move to that point. Under *all-trial* accounting it can. The difference matters. An early version of our own analysis used all-trial accounting and produced crossings between solver curves that disappeared once rejected trials were denied credit. We use accepted-only accounting throughout and report the other convention as a sensitivity check (section 5.11).

3.4 Primitive work as the unit of cost

Table 2 gives the median primitive work at matched tolerance (10^{-11}) over the 270 instances.

Four accepted Newton iterations use 5 EVDs and 24 Krylov iterations. SBB-Dual, Dykstra-APM and Anderson-APM use one EVD per iteration and no further primitive, although Anderson-APM’s least-squares update carries $O(mn^2)$ work that the EVD count does not record. Nine AGD-SDAJ-BH outer iterations use 70 EVDs, because each outer iteration contains $q = 2$ quasi-Newton inner

Table 3: Microseconds per EVD at matched tolerance (10^{-11}), $n = 100$, single-threaded BLAS.

solver	$\mu\text{s}/\text{EVD}$	relative to cheapest
SBB-Dual	1133	1.00
AGD-SDAJ-BH	1181	1.04
AGD-SDAJ	1188	1.05
Dykstra-APM	1278	1.13
Anderson-APM	1323	1.17
Newton-SIN	1341	1.18
Newton-SIN-BH	1695	1.50

steps, each with its own line search. A table of ‘iterations to convergence’ across these rows would compare four different quantities. We therefore report EVDs, Krylov iterations and line-search trials separately.

3.5 Timing protocol

All timings use Julia with `BLAS.set_num_threads(1)`. A warm-up solve is run and discarded, `GC.gc()` is called before the measured solve, and a single `@elapsed` surrounds the whole solve. A separate instrumented pass (`TimerOutputs`) gives the breakdown of time. In that breakdown, percentages are taken against the measured elapsed time, so any uninstrumented work shows up as ‘other’ time instead of inflating the EVD share. Timings were measured on all 270 instances at both the native ($10^{-7}n$) and the matched (10^{-11}) tolerance. The trajectory runs are a separate, untimed pass.

All seven variants were timed back to back in a single session, because timings from different sessions are not comparable: an earlier session on the same machine, with identical EVD counts on every run, measured the dual methods, whose code had not changed, between 3.7 and 4.2 times slower per EVD at $n = 100$. Each method is timed in an implementation that does not add work of its own. The projection methods rebuild the projection from the positive eigenpairs, as the dual methods do, rather than from all n , which removes an $O(n^3)$ product per EVD and gives bit-identical iterates on all 1104 runs checked; and Anderson-APM is timed in a form that updates its buffers in place, with iterates bit-identical to the transcription verified in section C. Timed as transcribed, it would have cost 2.3 to 3.8 times more per EVD. All timings were measured on one machine: Intel Core 7 150U (10 physical cores, 12 logical), 32 GB RAM, Windows 11 (build 26200), Julia 1.12.6 with OpenBLAS (ILP64) through libblastrampoline. The timing runs pin `BLAS.set_num_threads(1)`; the untimed trajectory pass at $n = 500$ was run with four BLAS threads, which changes wall-clock only and not the EVD counts it records.

3.6 Is the evd a fair unit of work?

The EVD count is a good proxy for cost only if one EVD costs about the same in every method. Table 3 and fig. 1 give the median wall-clock time per EVD at matched tolerance.

The spread is a factor of 1.50, and the largest bias works *against* Newton. Each Newton outer iteration includes a Krylov solve that no EVD count records, so counting EVDs understates Newton’s cost. Newton-SIN-BH is cheaper than SBB-Dual by a factor of 9.3 in EVDs (5 against 46.5) but by a factor of 5.4 in wall-clock time (0.009 s against 0.050 s). Anderson-APM’s least-squares update takes about 13% of its time at $n = 100$ and 15% at $n = 500$, so its EVDs cost 1.17 times SBB-Dual’s. On this family, then, EVD counts are biased in a known direction but preserve rank, and the bias

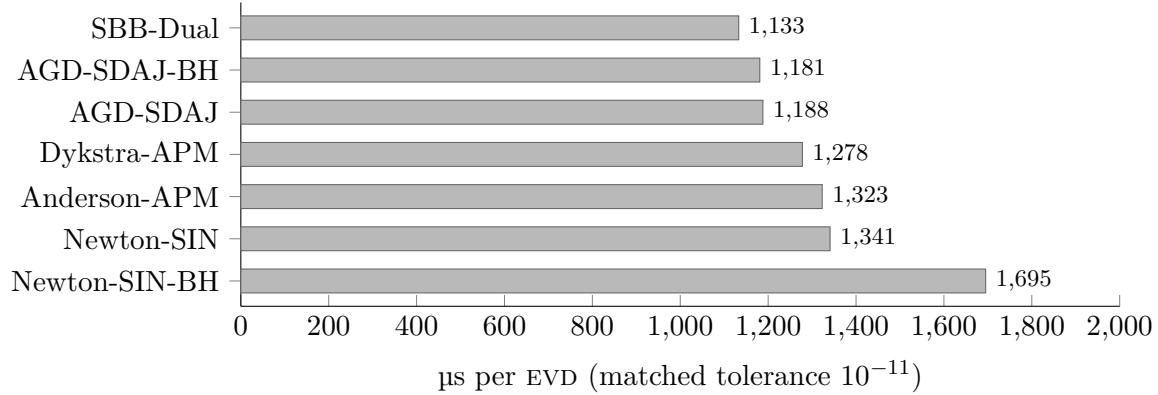


Figure 1: Median wall-clock time per EVD at $n = 100$, all seven variants timed in one session. The spread is a factor of 1.50 and works against Newton, whose EVDs are accompanied by Krylov solves that the EVD count does not record. Anderson-APM’s EVDs cost 1.17 times SBB-Dual’s, the difference being its least-squares update of $O(mn^2)$ work per EVD.

is small compared with the gaps being measured. We report both.

The cost metric does not change the ranking. At matched tolerance, ranking the seven timed variants by median EVDs and by median wall-clock time gives the same order:

$$\begin{aligned} \text{Newton-SIN-BH} &< \text{Newton-SIN} < \text{Anderson-APM} < \text{SBB-Dual} \\ &< \text{AGD-SDAJ-BH} < \text{AGD-SDAJ} < \text{Dykstra-APM}. \end{aligned}$$

This is a negative control for section 5. The reversals reported there come from the accuracy target and the instance regime. They are not artifacts of using the EVD as the unit of work. The same comparison at $n = 500$ gives a spread of 1.32, again with Newton-SIN-BH dearest per EVD, and again an identical ranking under the two cost metrics, so this negative control holds at both dimensions. It holds for medians; per instance the metric can decide a close pair, as for Anderson-APM against AGD-SDAJ-BH at $n = 500$, where Anderson-APM needs fewer EVDs on 13 of the 18 instances but less time on 11.

3.7 Summary of the protocol

1. Measure forward error against an exact X^* .
2. Run every solver to a common tight tolerance and record a trajectory point after every EVD.
3. Read costs off the trajectories at a common target; give rejected trials no accuracy credit.
4. Report EVDs, Krylov iterations, line-search trials and wall-clock time, never iterations alone.
5. Report the feasibility of the returned iterate ($\lambda_{\min}(X)$ and $\|\text{diag}(X) - e\|_{\infty}$), and state which iterate is returned.
6. Report reach rates beside every median that is computed over a subset of instances.

4 Two implementation traps

This section describes two implementation defects. Neither prevents convergence to the correct solution at the tolerances usually reported, so neither is visible from final errors. Both change measured cost substantially. We present them as evidence for the protocol of section 3. They are not new discoveries.

4.1 Trap 1: redundant Jacobian factorization

A semismooth Newton step at y_k needs the spectral decomposition of $C(y_k)$ to apply the generalized Jacobian [26]. That decomposition was already computed when $\theta(y_k)$ and $\nabla\theta(y_k)$ were evaluated, at the accepted trial of the previous line search. An implementation that recomputes it pays one extra EVD per outer iteration and gains nothing. On the instances where both variants converge, recomputing the decomposition raises Newton’s cost from 7 to 13 EVDs, an increase of about 86% (a factor of 1.86), with an identical final error. Every Newton variant in this paper caches the decomposition. The same class of defect turned up in our first reimplement of AGD-SDAJ (section B, defect 1). It was found in the same way, by counting primitives one at a time against a published count.

4.2 Trap 2: the finite-precision Armijo test

An objective-value sufficient-decrease test,

$$\theta(y_k + \alpha d_k) \leq \theta(y_k) + c \alpha \nabla\theta(y_k)^\top d_k, \quad (5)$$

fails in floating point once the predicted decrease $c \alpha |\nabla\theta(y_k)^\top d_k|$ falls below the rounding error in $\theta(y_k)$. Near the solution $|\nabla\theta^\top d| \approx \|\nabla\theta\|^2$, so the test becomes unresolvable at a gradient norm of roughly $\sqrt{u|\theta|/c}$, where u is the unit roundoff. From that point, no step can be accepted, even a step that improves both the gradient and the forward error. None of this is new. The cancellation problem is well known in general optimization [13]. For the NCM dual, Borsdorf [5, §4.7.1] derived the threshold (about 10^{-8} in $\|\nabla\theta\|$) and proposed alternatives, which became the modified step selection of Borsdorf and Higham [6, Sec. 3.3, Alg. 4.1]. Their Section 3.3 observes that when $|\phi(x)| = 1$, $\|p\|_2 < u^{1/2}$ and $\|\nabla\phi(x)\|_2 < u^{1/2}/2$, the computed values $fl(\phi(x+p))$ and $fl(\phi(x))$ coincide, so that the Armijo condition cannot be verified at all. Their Algorithm 4.1, Step 6, then tests whether the two objective values are *nearly equal*, using $|a-b| < \gamma u(1+|a|+|b|)$ with $\gamma = 100$, and if so accepts the full inexact Newton step provided $\|\nabla f(y_k + \alpha_k d_k)\|_2 / \|\nabla f(y_k)\|_2 \leq 1 - \mu$, falling back to a unit steepest-descent step otherwise. Our implementation uses their parameter values, $\mu_{\text{BH}} = 0.9$, $\rho = 0.5$ and $\sigma = 10^{-4}$ (we write μ_{BH} for their gradient-reduction parameter, since μ denotes the spectrum vector of proposition 2.1).

Semismooth Newton. With the naive test (5), Newton stalls on 56 of the 270 KKT instances, using 55,402 backtracks. With Borsdorf–Higham globalization (parameters $\mu = 0.9$, $\sigma = 10^{-4}$, halving, and $\gamma = 100$ in the near-equality test), it converges on all 270 with 8 backtracks. The count of 56 was reproduced exactly in a separate timing run on a separate code path: at matched tolerance, the naive-Armijo control exits at `max_iter` on 56/270. A predictor based on the ULP margin between the two sides of (5) separates stalling from non-stalling iterations with AUC 0.934. In all 2,181 affected iterations, the full step that was *rejected* would have reduced both the gradient norm and the exact forward error.

Table 4: AGD-SDAJ on `degen-r20-m1-d1e-10-p2`, as published and with the Borsdorf–Higham adaptation.

variant	outer	EVDs	exit	final $\ \nabla\theta\ $	$\ X - X^*\ _F$
AGD-SDAJ (as published)	115	4001	EVD budget	3.60×10^{-9}	6.09×10^{-9}
AGD-SDAJ-BH	10	67	converged	9.40×10^{-12}	1.82×10^{-11}

AGD-SDAJ. The same failure occurs in AGD-SDAJ [19], a separately published method with a different algorithmic structure. Line 5 of Algorithm 1 in that paper is the objective-value test (5). Run to the common tolerance 10^{-11} , AGD-SDAJ as published exhausts the 4000-EVD budget on 18 of the 270 instances. On instance `degen-r20-m1-d1e-10-p2`, the trajectory contains 3198 rejected monotone line-search trials, with θ and $\|\nabla\theta\|$ fixed at 6.0935×10^{-9} and 3.5986×10^{-9} . The predicted decrease is about 1.3×10^{-21} , well below the floating-point resolution of about 2×10^{-16} . We adapted the Borsdorf–Higham near-equality test to Algorithm 1’s step: when the objective difference cannot be resolved, a trial is accepted if it reduces $\|\nabla\theta\|$. With this change, all 18 stalls disappear (table 4). Across the family, the number of instances reaching the tolerance rises from 252/270 to 270/270, and the worst-case cost falls from 4035 EVDs to 131.

4.3 Scope of these findings

Three points limit what these results show.

1. *The phenomenon is known.* Borsdorf analysed it for the NCM dual [5, §4.7.1], and the fix was published in [6]. What we add is limited to (i) its recurrence in a separately published method that uses a naive Armijo test, where it is invisible under that method’s native stopping rule, and (ii) its quantification on the controlled family (56/270 and 18/270 stalls). For semismooth Newton the remedy is the published Borsdorf–Higham modified line search; for AGD-SDAJ it is our adaptation of their near-equality safeguard to the Algorithm 1 step of that method.
2. *It does not affect Huynh and Hwang’s reported results.* Their stopping rule, $\|\nabla\theta\|_2 \leq 10^{-7}n$, equals 10^{-5} on these instances and stops the method far above the precision floor at which the test fails. The stall appears only when the method is pushed to a tight common tolerance, which is what a matched-accuracy comparison requires. On their own instances, our reimplementations reproduce their published counts exactly for P8 and P9, and matches P7 in outer iterations and residual with one differing backtracking decision (section 7, section B).
3. *The adaptation is ours.* AGD-SDAJ-BH is our modification and is not specified by the original authors. We therefore report it alongside the unmodified method everywhere.

The general point is that whether a method appears to converge at a given tolerance can depend on a line-search detail that is unrelated to the algorithm’s design. A benchmark that stops each method under its own rule never exposes this.

5 Ranking under different conventions

5.1 Solvers

We compare six variants of five methods. All are implemented in Julia with a common evaluation kernel, and every evaluation of θ or $\nabla\theta$ costs exactly one EVD. Unless stated otherwise, each variant

starts from $y_0 = 0$ (Dykstra-APM starts from $X_0 = G$). The algorithms below are written at the same level of detail and use the notation $(\theta_k, g_k, X_k) \leftarrow \text{EVAL}(y_k)$ for one EVD of $C(y_k)$, with $X_k = \Pi_{\mathcal{S}_+^n}(C(y_k))$ and $g_k = \text{diag}(X_k) - e$. The stopping criterion in all algorithms is $\|g_k\|_2 \leq \tau$ (Dykstra-APM applies it to the diagonal of its PSD iterate). Here $\tau = 10^{-11}$ unless stated otherwise.

Newton-SIN-BH (algorithm 1) is the semismooth Newton method of Qi and Sun [26]. The Newton system is solved inexactly by conjugate gradients, using the block form of the generalized Jacobian restricted to the positive-eigenvalue columns (with $\Omega_{\beta\beta} = 0$) and a 10^{-12} diagonal shift. The spectral decomposition from the accepted trial is cached (section 4.1), and globalization uses the Borsdorf–Higham modified line search [6, Alg. 4.1, Step 6]. We adopt their globalization only: their Algorithm 4.1 also solves the Newton system with preconditioned MINRES rather than CG, which we do not adopt, so **Newton-SIN-BH** is not a reimplement of their algorithm. *Newton-SIN* is the same method with the naive Armijo test; it appears only as a control in sections 3 and 4.

AGD-SDAJ (algorithm 2) is Algorithm 4 of Huynh and Hwang [19], transcribed from the full text and checked against the published counts (section B). It uses $q = 2$ inner quasi-Newton steps with a diagonal secant preconditioner and the nonmonotone residual line search of their equation (6), followed by one accelerated gradient step with a monotone Armijo line search. *AGD-SDAJ-BH* is our adaptation of that Armijo step, described in section 4.2.

SBB-Dual (algorithm 3) is a spectral (Barzilai–Borwein) gradient method on (2). It uses the BB1 step, clamped to $[\eta, 2 - \eta]$ with $\eta = 0.01$, has no line search, performs one EVD per update, and finishes with a diagonal-rescaling step $\hat{X} = D^{-1/2} X D^{-1/2}$, $D = \text{Diag}(\text{diag } X)$, as in [6].

Dykstra-APM (algorithm 4) is Higham’s alternating projections with Dykstra’s correction [16]. Each iteration uses one EVD. The method returns the PSD half-iterate.

Anderson-APM (algorithm 5) is Dykstra-APM with Anderson acceleration, as given by Higham and Strabić [18]. The fixed-point map is one alternating-projections step on the pair $(Y, \Delta S)$, and the acceleration solves a least-squares problem over the last m differences of its residuals, taken here at the authors’ default $m = 2$. It costs one EVD per iteration plus $O(mn^2)$ work, and like Dykstra-APM it is scored on the PSD half-iterate. Our implementation transcribes their released code. Its iterates agree with an independent implementation to 2.3×10^{-12} , and it reproduces their published iteration counts where rounding at their stopping tolerance does not decide them: under the same eigensolver, both large-matrix counts exactly and 61 of 64 small ones, with the rest one iteration apart; on one ill-conditioned matrix the counts are decided by rounding and differ by up to 56 iterations (section C).

5.2 Design

All six variants were run on the 270 instances of section 2 at $n = 100$, to a common tolerance of 10^{-11} with a budget of 4000 EVDs, and a trajectory was recorded as described in section 3.2. We evaluated 11 conventions: the native exit; a common forward-error target $\|X - X^*\|_F \leq \varepsilon$ for $\varepsilon \in \{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}\}$; and a common dual residual $\|\nabla\theta\|_2 \leq \tau$ for the same five values. We also compared accepted-only with all-trial accounting. Section 5.6 repeats the study at $n = 500$.

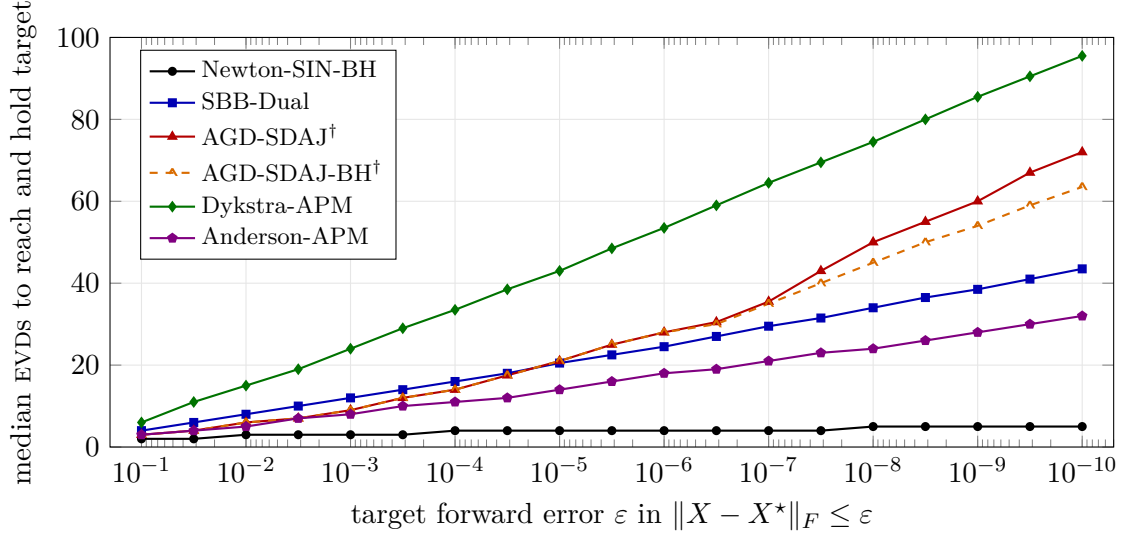


Figure 2: Median cumulative EVDs to reach and hold $\|X_k - X^*\|_F \leq \varepsilon$ (accepted-only accounting) over 270 instances, $n = 100$, on a half-decade grid. Anderson-APM is second at every target shown. SBB-Dual and AGD-SDAJ-BH cross between $\varepsilon \approx 3 \times 10^{-5}$ and 10^{-5} . † AGD-SDAJ-BH and every other solver reach and hold each target shown on all 270 instances. Unmodified AGD-SDAJ does so on all 270 down to 3×10^{-7} , and on 253/270 at 10^{-10} ; below 3×10^{-7} its medians are taken over the instances it reached. The curves stop at 10^{-10} . Cluster-bootstrap intervals for the decade targets are given in table 5; they are narrower than the plotted marks at most targets.

5.3 Cost of the semismooth Newton method across conventions

The 11 conventions produce three distinct orderings, and Newton-SIN-BH is cheapest in all three: it leads under every accuracy target, under both error measures, under either accounting rule and at every solver’s own exit, and it is strictly cheapest on every one of the 270 instances at 10^{-8} and 10^{-10} . In median, Anderson-APM is second under 10 of the 11 conventions and tied for second, with both AGD variants, under the loosest dual-residual target. Per instance its hold is much weaker: it is strictly cheapest after Newton-SIN-BH on 136 of the 270 instances at 10^{-8} and 139 at 10^{-10} , or on 157 and 142 counting ties. The conventions change cost ratios and the order of the remaining places; they never change which method is cheapest. The rest of this section is about those remaining places, which is where the choice of test set decides the answer.

5.4 Dependence of the ordering on the accuracy target

Table 5 and fig. 2 show median cost to a common target. Anderson-APM is second at every target, from 5 EVDs at 10^{-2} to 32 at 10^{-10} , where AGD-SDAJ-BH needs 63.5 and SBB-Dual 43.5. The places below it move. At loose targets (10^{-2} , 10^{-4}), both AGD variants are cheaper than SBB-Dual. From 10^{-6} onward, SBB-Dual is cheaper. The same reversal occurs at the same point under the forward-error and the dual-residual criteria, so it does not depend on which error measure is used. A paper that reports at a loose tolerance would place AGD-SDAJ above SBB-Dual, and a paper that reports at a tight tolerance would place them the other way. Both are defensible one-number summaries of the same runs.

Table 5: Median EVDs to reach and hold $\|X - X^*\|_F \leq \varepsilon$ (accepted-only) at $n = 100$, with cluster-bootstrap 95% intervals in brackets ($B = 2000$, resampling the five paired seeds within each rank, so that all 18 (m, δ) cells of a drawn seed move together). Rows are ordered by the 10^{-8} column. Entries marked * are medians over the subset of instances that reached the target, the subset size in small type; all other entries are over 270/270.

solver	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
Newton-SIN-BH	3 [3, 3]	4 [3, 4]	4 [4, 4]	5 [5, 5]	5 [5, 5]
Anderson-APM	5 [5, 5]	11 [11, 11]	18 [17, 18.5]	24 [24, 26]	32 [30, 34]
SBB-Dual	8 [8, 8]	16 [15.5, 16]	24.5 [24.5, 24.5]	34 [34, 34]	43.5 [43, 43.5]
AGD-SDAJ-BH	6 [6, 6.5]	14 [13, 15]	28 [27, 29]	45 [42, 48.5]	63.5 [62, 66]
AGD-SDAJ	6 [6, 6.5]	14 [13, 15]	28 [27, 29]	50* [46, 52] [264]	72* [70, 75] [253]
Dykstra-APM	15 [15, 15]	33.5 [33.5, 33.5]	53.5 [53.5, 53.5]	74.5 [74.5, 74.5]	95.5 [95.5, 95.5]

Reach and hold. A target counts as reached at the first accepted iterate from which the error stays at or below the target for the rest of the run. The accelerated methods are not monotone in forward error, so this differs from the first iterate at which the error dips below the target; scoring the first dip alone would understate their cost. Under this rule Newton-SIN-BH, AGD-SDAJ-BH, SBB-Dual and Dykstra-APM reach every target on all 270 instances. Only unmodified AGD-SDAJ misses some: 264/270 and 253/270 at forward targets 10^{-8} and 10^{-10} (264 and 254 for the residual), which is consistent with its 18 budget-exhausting stalls (section 4.2). Its entries marked * are medians over the instances it reached, and are therefore optimistic. The second/third-place reversal is the same whether or not those entries are included.

Uncertainty. The bootstrap intervals in table 5 are narrow and do not overlap between SBB-Dual and the AGD variants at any target, so the change of order between 10^{-4} and 10^{-6} is not a sampling artefact.

The paired per-instance picture at $n = 100$ is, however, much closer than the aggregate medians suggest. Against AGD-SDAJ-BH at $\varepsilon = 10^{-2}$ the median paired difference is only +2 EVDS (95% CI [1, 2]), SBB-Dual is cheaper on 40.4% of the 270 instances, and 10 instances tie. The $n = 100$ ordering between these two is therefore a modest average effect with substantial per-instance overlap, in contrast to $n = 500$ (section 5.6), where SBB-Dual is cheaper on 0 of 270 instances at the same target.

5.5 Dependence of the ordering on the degeneracy parameters

Aggregate medians can hide dependence on the regime. At a fixed target (forward error 10^{-8}), the 54 (r, m, δ) cells produce 10 distinct orderings (table 6).

Newton-SIN-BH is cheapest in every cell. Below it, at $n = 100$, the order is governed almost entirely by the rank r of the solution. AGD-SDAJ-BH is cheaper than SBB-Dual in all 18 cells with $r = 5$; SBB-Dual is cheaper in 16 of the 18 cells with $r = 20$ and in all 18 with $r = 50$. Within a rank, the near-zero multiplicity m and the separation δ rarely change this pair’s order. Both AGD variants fall below Dykstra-APM to last place in 14 cells: all 12 cells with $r = 50$ and $m \in \{5, 20\}$, and the two cells with $r = 20$, $m = 20$ and $\delta \in \{10^{-8}, 10^{-10}\}$. This block is determined by rank and multiplicity; the separation plays no visible role in it. The instance family was built to vary these quantities one at a time. On a collection of real matrices with uncontrolled spectra, the ordering could not be associated with rank in this way.

Table 6: Orderings by cell at forward target 10^{-8} (54 cells, five paired instances each), with the number of cells of each rank producing each ordering.

ordering (cheapest first)	cells	$r=5$	$r=20$	$r=50$
Newton-SIN-BH < Anderson-APM < AGD-SDAJ-BH < AGD-SDAJ < SBB-Dual < Dykstra-APM	16	15	1	0
Newton-SIN-BH < SBB-Dual < Anderson-APM < Dykstra-APM < AGD-SDAJ-BH < AGD-SDAJ	13	0	2	11
Newton-SIN-BH < Anderson-APM < SBB-Dual < AGD-SDAJ-BH < AGD-SDAJ < Dykstra-APM	12	0	12	0
Newton-SIN-BH < SBB-Dual < Anderson-APM < AGD-SDAJ-BH < AGD-SDAJ < Dykstra-APM	7	0	1	6
Newton-SIN-BH < Anderson-APM < AGD-SDAJ-BH < SBB-Dual < AGD-SDAJ < Dykstra-APM	1	0	1	0
Newton-SIN-BH < SBB-Dual < Anderson-APM < AGD-SDAJ-BH < Dykstra-APM < AGD-SDAJ	1	0	1	0
Newton-SIN-BH < AGD-SDAJ < AGD-SDAJ-BH < Anderson-APM < SBB-Dual < Dykstra-APM	1	1	0	0
Newton-SIN-BH < AGD-SDAJ-BH < AGD-SDAJ < Anderson-APM < SBB-Dual < Dykstra-APM	1	1	0	0
Newton-SIN-BH < AGD-SDAJ-BH < Anderson-APM < AGD-SDAJ < SBB-Dual < Dykstra-APM	1	1	0	0
Newton-SIN-BH < SBB-Dual < Anderson-APM < Dykstra-APM < AGD-SDAJ < AGD-SDAJ-BH	1	0	0	1

A crossover between published methods. The reversal is not confined to SBB-Dual. Dykstra-APM and AGD-SDAJ-BH, both published methods, exchange places over the same rank range. At $r = 5$ AGD-SDAJ-BH is cheaper on all 90 instances at 10^{-8} (median difference +223.5 EVDs) and at $r = 20$ on 80 of 90, with one tie; at $r = 50$ Dykstra-APM is cheaper on 61 of 90 (median -3.5 , cluster-bootstrap 95% interval $[-6, -2]$) and on 75 of 90 at 10^{-10} (-8.5 , $[-12, -5]$). All five seeds agree on the sign at $r = 5$ and at $r = 50$. The direction at $r = 50$ is resolved but the margin is small, about an eighth of AGD-SDAJ-BH’s median cost of 28 EVDs, and 26 of the 90 instances go the other way. The reversal is therefore reproducible among previously published methods, and it moves with dimension in the same direction: at $n = 500$ AGD-SDAJ-BH is still ahead at $r = 150$ (Dykstra-APM cheaper on 13 of 45 at 10^{-10} , section 5.7), and on the equity panel at $n = 2105$ the corresponding pair is Anderson-APM and AGD-SDAJ-BH, which exchange places between $r = 200$ and $r = 400$ (section 5.9).

The location of the crossover is nevertheless specific to $n = 100$. At $n = 500$ (sections 5.6 and 5.7) SBB-Dual is behind at every rank up to $r = 50$ and ahead from $r = 75$: both the direction of the effect and the crossover survive, but the crossover moves to a higher rank.

That rankings depend on the required accuracy is expected in principle [25, 15]. What the controlled family adds is that the variation is organized by a single spectral quantity which the construction sets directly.

Each cell holds five instances, so the count of distinct orderings overstates how much genuinely varies. Resampling the paired seeds of each cell ($B = 2000$) leaves the adjacent pair AGD-SDAJ-BH against AGD-SDAJ unresolved in most cells, as expected of two methods differing only in a safeguard that rarely activates at $n = 100$; reading unresolved pairs as ties collapses much of table 6. What survives is the placement of SBB-Dual relative to the AGD pair, resolved in the great majority of cells. Figure 3 shows the same comparison cell by cell as a ratio of median

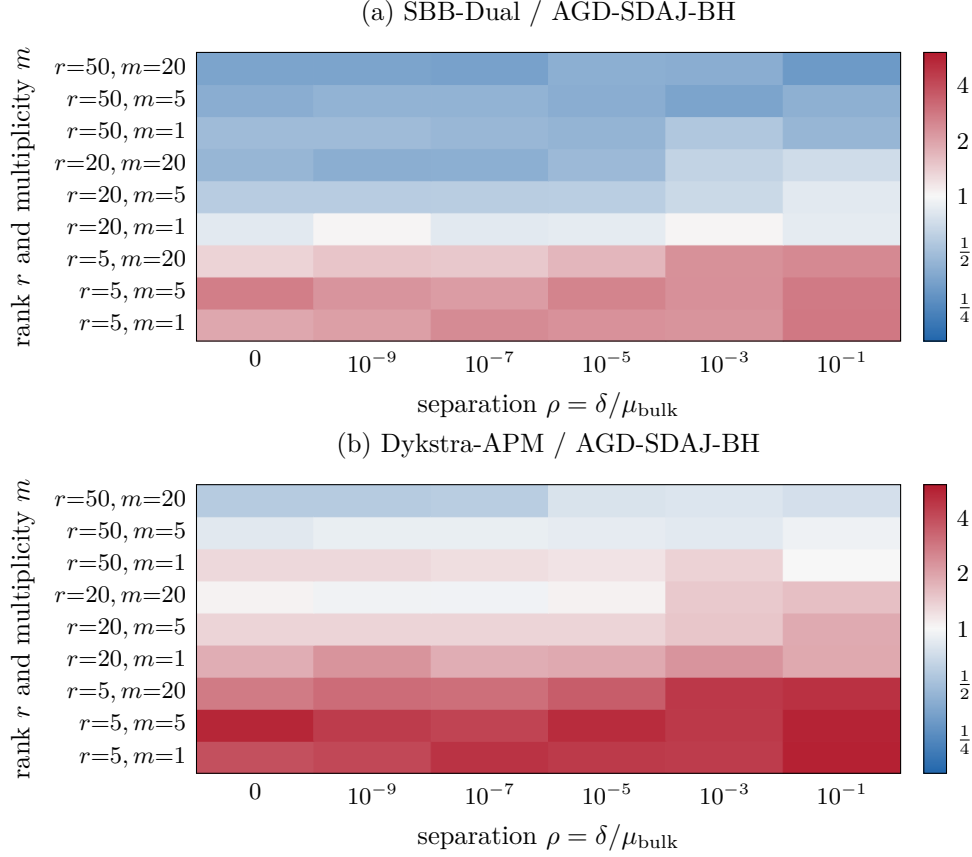


Figure 3: Ratio of median EVDs to reach and hold $\|X - X^*\|_F \leq 10^{-8}$ for each of the 54 cells at $n = 100$ (five paired instances per cell), on one shared logarithmic colour scale: red cells favour AGD-SDAJ-BH, blue cells the other method, white is parity. (a) SBB-Dual over AGD-SDAJ-BH. The ratio is organized primarily by rank: it exceeds 1 in every $r = 5$ cell and is below 1 in every $r = 50$ cell. The near-zero multiplicity m modulates the level within a rank, most visibly at $r = 20$ (from 1.03 at $m = 1$ to 0.40 at $m = 20$), whereas the separation ρ has little effect. (b) Dykstra-APM over AGD-SDAJ-BH, two published methods. The organization is the same, because Dykstra-APM costs between 1.6 and 2.7 times SBB-Dual in every cell, so the whole map is shifted towards AGD-SDAJ-BH: the ratio lies between 2.7 and 5.7 at $r = 5$, between 0.94 and 2.2 at $r = 20$ and between 0.58 and 1.35 at $r = 50$, and it falls below 1 in 14 cells, all 12 with $r = 50$ and $m \in \{5, 20\}$ and two with $r = 20$ and $m = 20$.

costs, which separates the three controlled quantities: rank sets the level, the near-zero multiplicity modulates it within a rank, and the separation has almost no effect.

5.6 The same design at $n = 500$

The instance family was regenerated at $n = 500$ with the same grid. All 270 instances screened successfully; the acceptance rate at $r = 5$ fell to 5 of 7 candidate seeds, the first rank to show rejections, which is what proposition 2.4 predicts. The ranking study uses all five paired seeds of each rank, 270 instances, with caps of 1200 iterations for SBB-Dual and 800 for Dykstra-APM. It was run in two parts, seeds 0–2 (about 8.6 h) and seeds 3–4 (about 7.1 h), with identical code and settings.

Table 7: $n = 500$, 270 instances. Median EVDs to reach and hold $\|X - X^*\|_F \leq \varepsilon$, with cluster-bootstrap 95% intervals ($B = 2000$, resampling paired seeds within rank) and reach counts where these are below 270.

solver	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
Newton-SIN-BH	3 [3, 3]	4 [3, 4]	4 [4, 4]	5 [5, 5]	5 [5, 5]
Anderson-APM	13 [12, 14.5]	23.5 [23, 28]	40.5 [36.5, 43.5]	53.5 [53, 55]	70.5 [67.5, 74]
AGD-SDAJ-BH	9 [9, 9]	30 [29, 32]	53.5 [50.5, 56]	79 [76, 80]	101.5 [98.5, 105]
AGD-SDAJ	9 [9, 9]	30 [29, 32]	55.5 [52, 58] ^[268]	87 [84, 89] ^[259]	119 [117, 123] ^[242]
SBB-Dual	44 [44, 44]	97 [97, 97]	152 [152, 152]	207.5 [207, 208]	260 [259, 262] ^[261]
Dykstra-APM [‡]	87 [86.5, 88]	195 [194.5, 195.5]	191 [191, 192] ^[180]	258 [258, 260] ^[180]	325.5 [325, 327.5] ^[180]

On this grid ($r \leq 50$), two of the $n = 100$ conclusions survive and two do not; section 5.7 shows that the two that fail return at higher rank.

Confirmed. Newton-SIN-BH is again cheapest at every target, at 5 EVDs. The stall of unmodified AGD-SDAJ recurs (31 of 270 runs exhaust the EVD budget, against none for AGD-SDAJ-BH, whose worst case is 240 EVDs against 4039).

Not reproduced for $r \leq 50$. The change of order between SBB-Dual and the AGD variants seen in table 5 is absent: at $n = 500$ AGD-SDAJ-BH is ahead of SBB-Dual at all five targets. Only Anderson-APM moves, from fourth at 10^{-2} to second from 10^{-4} on (table 7). The paired per-instance comparison is one-sided. SBB-Dual is cheaper on 0 of 270 instances at 10^{-2} , 3 at 10^{-4} and 15 at 10^{-8} . We report these counts and the cluster-bootstrap intervals of table 7 rather than an instance-level significance test: the design reuses each drawn U across all 18 (m, δ) cells of its rank, so the 270 comparisons form 15 seed clusters, and a test treating them as independent would overstate the evidence by a wide margin. The rank effect of section 5.5 keeps its direction but no longer produces a crossover: the median paired difference at 10^{-8} falls from +669.5 EVDs at $r = 5$ to +129 at $r = 20$ and +13 at $r = 50$, where SBB-Dual is cheaper on 17% of instances. Cell by cell, SBB-Dual is ahead of AGD-SDAJ-BH in 0 of the 54 cells with a resolved bootstrap interval (47 resolved in favour of AGD-SDAJ-BH, 7 unresolved). SBB-Dual is thus behind at every rank tested at $n = 500$, while it led at $r \geq 20$ at $n = 100$. The shrinking gap, however, points to a crossover beyond the grid, which section 5.7 confirms.

This is not an artifact of the caps. SBB-Dual converged on all 270 instances (worst case 1075 EVDs against its 1200 cap) and reached every target except 10^{-10} on nine instances. [‡]Dykstra-APM, by contrast, *is* censored: 90 of 270 runs stop at the 800-iteration cap, and from 10^{-6} down its medians cover only 180 instances. Its position at $n = 500$ should be read as censored rather than measured.

The scaling is the cleaner way to state the difference. Between $n = 100$ and $n = 500$ the median cost to 10^{-8} rises by a factor of 6.1 for SBB-Dual (34 to 207.5 EVDs) and by 1.8 for AGD-SDAJ-BH (45 to 79), while Newton-SIN-BH is unchanged at 5. The order below first place therefore reflects the test set, including its dimension, rather than a fixed property of the methods — which is the paper’s point, in a stronger form than the $n = 100$ data alone supports.

5.7 Higher ranks at $n = 500$

The grid of section 2 stops at $r = 50$. To locate the crossover we generated 135 further instances at $n = 500$ with $r \in \{75, 100, 150\}$, $m \in \{1, 5, 20\}$, $\delta \in \{10^{-4}, 10^{-8}, 0\}$, $\mu_{\text{bulk}} = 0.1$ and five paired seeds per rank, all accepted by screening, and ran every variant except unmodified AGD-SDAJ (table 8).

Table 8: $n = 500$, higher ranks, 45 instances per rank. Median EVDs to reach and hold $\|X - X^*\|_F \leq 10^{-8}$; instances (out of 45) on which SBB-Dual is cheaper than AGD-SDAJ-BH at three targets; and the median paired difference SBB-Dual minus AGD-SDAJ-BH at 10^{-8} with a 95% interval from resampling the five seeds ($B = 2000$). All runs converged.

r	median EVDs to 10^{-8}					cheaper at 10^{-8} , of 45	
	Newton	Anderson	SBB-Dual	AGD-SDAJ-BH	Dykstra	SBB vs AGD	Anderson vs AGD
75	5	26	51	60	110	34	45
100	4	22	37	52	81	43	45
150	5	18	22	41	52	45	45

Between $r = 50$, where SBB-Dual is behind by 13 EVDs at 10^{-8} (section 5.6), and $r = 75$, where it is ahead by 10, the order reverses. The dependence on the accuracy target returns with it: at $r = 75$ and $r = 100$ the two methods are level at 10^{-4} (median differences +1 and 0), with AGD-SDAJ-BH ahead for $m = 1$, while SBB-Dual is ahead at 10^{-8} and 10^{-10} . Both patterns of section 5.5 therefore survive at $n = 500$; what changes with dimension is the rank at which they appear, from between $r = 5$ and $r = 20$ at $n = 100$ to between $r = 50$ and $r = 75$ at $n = 500$. We report these as brackets and fit no trend to them: three dimensions with three or four grid points each cannot separate proportional growth from any other increasing rule, and the equity instances of sections 5.8 and 5.9 sit outside the synthetic family in more ways than dimension. Newton-SIN-BH is strictly cheapest on all 135 instances at every target from 10^{-4} to 10^{-10} .

A pilot of 48 instances at $r \in \{100, 150\}$, repeated with $\mu_{\text{bulk}} = 0.01$ on the same X^* and hence a tenfold smaller violation, gave the same result: at 10^{-8} the median paired difference was -12.5 against -14 EVDs at $r = 100$ and -19.5 against -18 at $r = 150$. The effect follows the rank of the solution, not the size of W .

5.8 Instances built from equity returns

Two further test sets start from one real correlation matrix: the sample correlation matrix C of daily log returns, computed from split-adjusted closing prices, of 550 US-traded equities over 1397 trading days (January 2020 to July 2025), with no missing values. The panel holds 50 names under each of the eleven sector labels of its source files and includes some ADRs and one closed-end fund. C is a valid correlation matrix, with $\lambda_{\min}(C) = 0.019$. It serves here only as a realistic input, not as an estimate of market structure; one series (CHRD) contains a corporate-action splice, which does not affect that use.

KKT instances seeded by the data. Because C has full rank it cannot itself be the solution of a nontrivial instance of section 2, whose construction needs a kernel. Its rank- r factor model can: with $C \approx V_r \Lambda_r V_r^\top$, set $U = V_r \Lambda_r^{1/2}$, scale its rows to unit norm and take $X^* = UU^\top$. The construction of section 2 then gives G with X^* as its *exact* nearest correlation matrix, whose eigenvectors come from the market data rather than from a random U . Replicates differ by a Haar-random rotation of the kernel basis. Two settings differ from the synthetic family, both forced by screening: at $r = 5$ the factor model misses 95% of the off-diagonal correlation and drives some $|X_{ij}^*|$ to 0.99995, so every $r = 5$ instance fails $|G_{ij}| \leq 1$, and with $\mu_{\text{bulk}} = 0.1$ every rank fails it. We therefore use $r \in \{20, 50, 100\}$ (off-diagonal relative fit 0.59, 0.39, 0.24) and $\mu_{\text{bulk}} = 0.01$, with $\delta \in \{10^{-5}, 10^{-9}, 0\}$ so that $\text{sep} = \delta/\mu_{\text{bulk}}$ stays on the synthetic grid, $m \in \{1, 5, 20\}$ and three rotations: 81 instances. All pass screening, with worst exactness 1.1×10^{-12} . They are only mildly

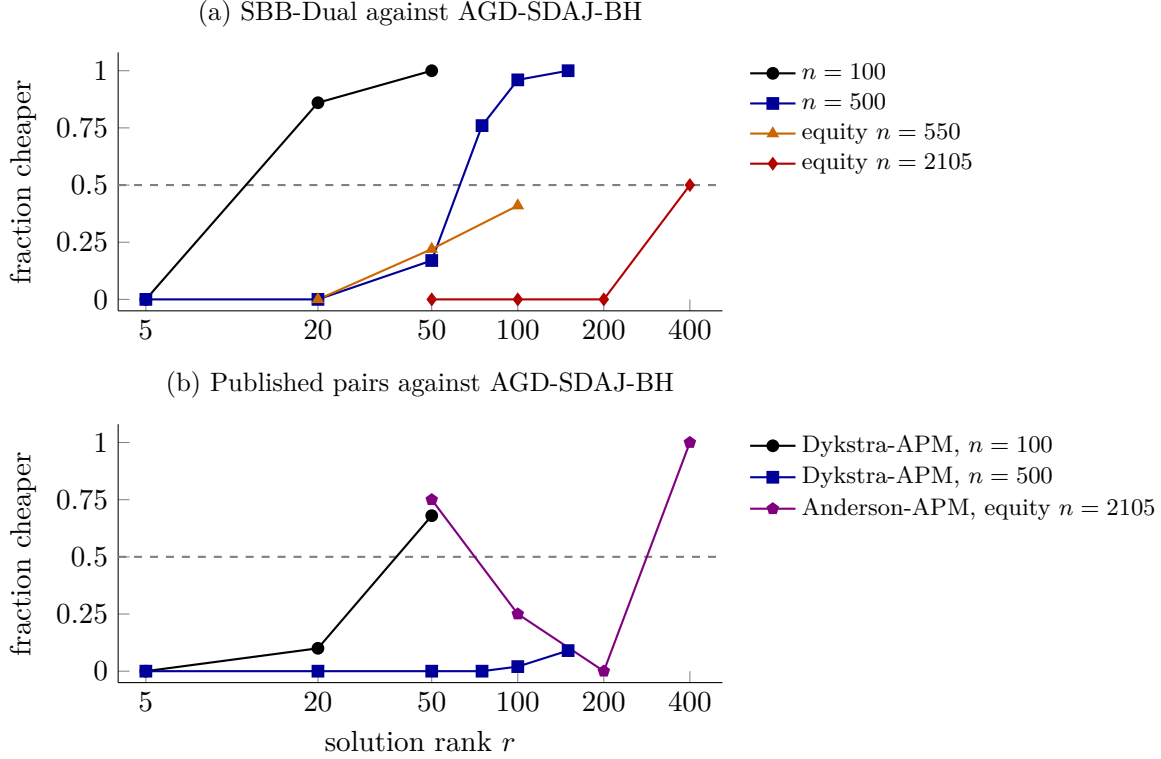


Figure 4: Where pairs of methods exchange places. Each point is the fraction of instances at one rank on which the named method reaches $\|X - X^*\|_F \leq 10^{-8}$ in fewer EVDS than AGD-SDAJ-BH; a run that never reaches the target counts as the more expensive one, and the dashed line marks parity. (a) SBB-Dual: the exchange happens between $r = 5$ and $r = 20$ at $n = 100$, between $r = 50$ and $r = 75$ at $n = 500$, and has not been reached by $r = 100$ on the equity instances at $n = 550$ or by $r = 200$ at $n = 2105$; the rightmost point rests on four instances and its ordering reverses at 10^{-10} (section 5.9). (b) Two published methods: Dykstra-APM overtakes AGD-SDAJ-BH between $r = 20$ and $r = 50$ at $n = 100$ but not by $r = 150$ at $n = 500$. At $n = 2105$, where each point rests on four instances, Anderson-APM is ahead at $r = 50$, behind at $r = 100$ and $r = 200$, and ahead again at $r = 400$, so its order is not monotone in rank. For SBB-Dual and for Dykstra-APM the rank at which the exchange occurs rises with dimension; it is reported as a bracket per dimension and no trend is fitted to it.

invalid, $\lambda_{\min}(G) \in [-4.6, -1.2] \times 10^{-3}$.

Anderson-APM is again second at every rank (108, 64 and 53 EVDS to 10^{-8}), and it is cheaper than AGD-SDAJ-BH on 15, 25 and 23 of the 27 instances at $r = 20$, 50 and 100. The margin therefore widens with rank here too, on matrices whose eigenvectors come from market data rather than from a random orthogonal basis.

Table 9 repeats the pattern of the synthetic family on data-derived spectra. Newton-SIN-BH is cheapest on all 81 instances at every target. Unmodified AGD-SDAJ exhausts its 4000-EVD budget on 10 of 81, and AGD-SDAJ-BH on none. Rank again organizes the comparison between SBB-Dual and AGD-SDAJ-BH: the median paired difference at 10^{-8} falls from +128 EVDS at $r = 20$ to +17 at $r = 50$ and +7 at $r = 100$. At $r = 100$ the gap nearly closes: SBB-Dual is cheaper on 11 of 27 instances at 10^{-8} and 13 of 27 at 10^{-10} , ahead on all nine with $m = 1$ and behind on all nine with $m = 20$. On synthetic instances of similar size SBB-Dual is already ahead at $r = 75$ (section 5.7),

Table 9: KKT instances seeded by equity returns, $n = 550$, 27 instances per rank. Median EVDs to reach and hold $\|X - X^*\|_F \leq 10^{-8}$ against the exact X^* ; bold marks the cheapest, and a bracketed count is the number of instances that reached the target where fewer than 27 did. Out of 27 instances, SBB-Dual is cheaper than AGD-SDAJ-BH on 0, 6 and 11 at $r = 20, 50$ and 100 (0, 6 and 13 at 10^{-10}), and Anderson-APM on 15, 25 and 23 (15, 25 and 24 at 10^{-10}).

r	Newton-SIN-BH	Anderson-APM	AGD-SDAJ-BH	AGD-SDAJ	SBB-Dual	Dykstra-APM
20	4	108	117	154	249	500
50	4	64	92	94.5 [24]	112	228
100	4	53	65	79 [25]	72	148

Table 10: Perturbed equity correlation matrix, $n = 550$, five draws per σ . Median EVDs to reach and hold forward error 10^{-8} against the computed reference, and the number of draws (out of 5) on which SBB-Dual is cheaper than AGD-SDAJ-BH.

σ	Newton-SIN-BH	SBB-Dual	Anderson-APM	Dykstra-APM	AGD-SDAJ-BH	AGD-SDAJ	SBB cheaper
0.005	5	10	11	17	20	24	5
0.01	5	15	18	30	20	30	5
0.03	6	29	31	60	29	29	2 (1 tie)
0.1	7	67	59	140	38	42.5 [4]	0

so eigenvectors taken from market data appear to move the crossover to a still higher rank. The replicates share X^* within a rank, so these are counts, not independent samples.

Perturbed data matrix. The second set follows test problem P12 of [19]: $G = \text{clip}(C + \sigma E, -1, 1)$ with unit diagonal, where E is symmetric with standard normal off-diagonal entries, for $\sigma \in \{0.005, 0.01, 0.03, 0.1\}$ and five draws each. The smallest eigenvalue of G falls from about -0.09 to -4.2 and the number of negative eigenvalues rises from about 60 to 240; at $\sigma = 0.1$ about 0.02% of entries are clipped. No exact solution exists, so forward error is measured against a Newton reference with $\|\nabla\theta\|_2 \leq 1.2 \times 10^{-13}$ on every matrix.

Here the degree of invalidity reverses the order below first place (table 10); the most negative eigenvalue and the number of negative eigenvalues grow together with σ , so the two cannot be separated. For small perturbations SBB-Dual is second, 1.3 to 2 times cheaper than AGD-SDAJ-BH in median, and at $\sigma = 0.005$ even Dykstra-APM is ahead of both AGD variants; at $\sigma = 0.03$ the two are level; at $\sigma = 0.1$ AGD-SDAJ-BH is cheaper on every draw, by a median factor of about 1.7. The same reversal holds at 10^{-4} and 10^{-6} . It is consistent with the matrices of section 5.10, where the mildly invalid Rocky Mountain matrix favours SBB-Dual and the larger, more strongly invalid matrices favour AGD-SDAJ-BH. Newton-SIN-BH is cheapest on every matrix at every target from 10^{-4} to 10^{-8} , and unmodified AGD-SDAJ stalls once, at $\sigma = 0.1$.

5.9 An equity panel at $n = 2105$

The panel of section 5.8 is one of 550 names over 1397 days. To test the same questions an order of magnitude higher we built a second panel of 2105 US-traded common stocks over 1253 trading days (September 2021 to September 2026), selected from an exchange listing by complete price history, a median traded value above \$1 M per day and one share class per company. Here n exceeds the sample length, so the sample correlation matrix C is singular: 853 of its eigenvalues are zero to working precision, and C sits exactly on the boundary of the feasible set.

Table 11: Equity panel at $n = 2105$, four instances per rank. Median EVDs to reach and hold $\|X - X^*\|_F \leq 10^{-8}$ against the exact X^* , and the number of instances on which SBB-Dual is cheaper than AGD-SDAJ-BH at 10^{-8} and at 10^{-10} . A dash marks a method that did not reach the target within its cap of 800 EVDs.

r	Newton-SIN-BH	Anderson-APM	AGD-SDAJ-BH	AGD-SDAJ	SBB-Dual	Dykstra-APM	SBB cheaper
50	5	147.5	159.5	201.5	650	–	0, 0
100	5	233	143	179	442	–	0, 0
200	5	158.5	111	116.5	244	491	0, 0
400	5	78	109.5	130	109	222.5	2, 1

Two sets were generated from it. The perturbed set adds σE as in section 5.8 for $\sigma \in \{0.005, 0.01, 0.03, 0.1\}$ with two draws each; the resulting matrices are strongly invalid, with $\lambda_{\min}(G)$ from -0.34 to -8.7 and 599 to 968 negative eigenvalues. The KKT set uses the rank- r factor model of C as X^* with $r \in \{50, 100, 200, 400\}$, $m \in \{1, 20\}$, $\delta = 0$ and two kernel rotations, 16 instances with exact solutions. At this size $\mu_{\text{bulk}} = 0.1$ passes the screen at every rank, so these instances use the same bulk level as the synthetic family.

Newton-SIN-BH takes 5 EVDs on every instance, as at $n = 100$ and $n = 500$: the cost of the leading method is flat in n over a twenty-fold range, while the first-order methods grow by an order of magnitude. The gap between SBB-Dual and AGD-SDAJ-BH again narrows with rank, from +490 EVDs at $r = 50$ to -0.5 at $r = 400$ at the 10^{-8} target. We do not claim a crossover at $r = 400$: the median there is -0.5 EVDs with a cluster interval of $[-27, +21.5]$, and at 10^{-10} the sign turns over (+8, with AGD-SDAJ-BH cheaper on three of the four instances). The crossover is therefore not reached by $r = 200$ and is unresolved at $r = 400$.

The two rotations per cell are weak replicates for several methods. Newton-SIN-BH’s costs are identical across rotations in every cell and SBB-Dual’s in six of eight; only the AGD variants vary with the rotation throughout. At $r = 200$ and $r = 400$ the costs of Anderson-APM and Dykstra-APM also differ between rotations by at most two EVDs, although Anderson-APM’s do vary at $r = 50$ and $r = 100$.

Anderson-APM converges on all 16 instances, and with AGD-SDAJ-BH it forms the one pair of published methods here whose order changes with rank. It is cheapest after Newton-SIN-BH at $r = 50$ and $r = 400$, but AGD-SDAJ-BH is cheaper at $r = 100$ and $r = 200$. At 10^{-8} and 10^{-10} AGD-SDAJ-BH is cheaper on 4 of 4 instances at $r = 200$, by 33 to 70 EVDs, and Anderson-APM on 4 of 4 at $r = 400$, by 6 to 68; because Anderson-APM is nearly rotation-invariant at these ranks, each count rests on two distinct instances, one per multiplicity. Near-zero multiplicity also matters here in a way it did not at $n = 100$: at $r = 100$ the two instances with $m = 20$ favour AGD-SDAJ-BH by about 190 EVDs, while those with $m = 1$ favour Anderson-APM or tie. With four instances per rank these are descriptive counts, not estimates.

Dykstra-APM was run on the same 16 instances at a cap of 800 EVDs, one EVD costing about 1.8s at this size. It is last at every rank and target it reaches, so it changes no ordering, but its own behaviour across the set is the sharpest rank dependence we observed. At $r = 50$ and $r = 100$ it exhausts the cap, ending at forward errors of 8.8×10^{-6} and 6.7×10^{-8} ; at $r = 200$ and $r = 400$ it converges on its own diagonal-feasibility test, in 647 to 662 and 301 to 305 EVDs. Its cost to a loose target falls monotonically with rank, from 556 EVDs at $r = 50$ to 93 at $r = 400$ for $\varepsilon = 10^{-4}$, a sixfold drop within one test set, while Newton-SIN-BH stays at 4. A study drawing its instances only from low-rank solutions would report Dykstra-APM as a method that does not converge at this size; one drawing from high-rank solutions would report it reaching 10^{-10} in 289 EVDs.

Table 12: Perturbed equity panel at $n = 2105$, two draws per σ . EVDs to reach and hold forward error 10^{-8} against a computed reference with $\|\nabla\theta\|_2 \leq 3.2 \times 10^{-13}$.

σ	Newton-SIN-BH	SBB-Dual	AGD-SDAJ-BH	AGD-SDAJ
0.005	6, 6	19, 19	34, 22	34, 76
0.01	6, 6	29, 28	32, 29	33, 34
0.03	7, 7	71, 67	37, 44	38, 48
0.1	8, 8	165, 162	68, 66	47, 72

The perturbed set reproduces the pattern of table 10 at four times the dimension (table 12): SBB-Dual is second at the two smallest σ and about twice as expensive as AGD-SDAJ-BH at $\sigma = 0.1$, with the change of order between $\sigma = 0.01$ and $\sigma = 0.03$ in both panels. Unmodified AGD-SDAJ exhausted its 800-EVD budget on two of the 16 KKT instances, both at $r = 400$, where AGD-SDAJ-BH converged in 119 and 140 EVDs; this is the stall of section 4.2 at the largest size we tested.

Where Anderson-APM is not second. Its second place is a property of the KKT families, not of the method. On the perturbed equity matrices it is third at every σ , cheaper than AGD-SDAJ-BH on all five draws at $\sigma \in \{0.005, 0.01\}$ and on none at $\sigma \in \{0.03, 0.1\}$, so the degree of invalidity reorders it exactly as it reorders SBB-Dual (table 10). On the four literature matrices it is second on the Rocky Mountain matrix (6 EVDs), but third on **cor1399** (103 against 66 for AGD-SDAJ-BH) and fourth on **cor3120** (155 against 41), the two matrices where every projection method is slow (table 13). A comparison drawn only from the exact families would therefore overstate its standing, consistent with the regime dependence reported throughout this section.

5.10 Four matrices from the literature

The family of section 2 is synthetic. To check that the regime dependence above is not an artifact of the construction, we ran the same protocol on four invalid correlation matrices from the literature: the Rocky Mountain carbon-storage matrix ($n = 94$), **cor1399** ($n = 1399$), **cor3120** ($n = 3120$) and **bccd16** ($n = 3250$), all from the Anymatrix **CORRINV** collection [17] used by [19].

No exact solution is available here, which is the situation the rest of this paper avoids. Forward error is measured against a reference computed by semismooth Newton driven towards 10^{-13} ; for **cor1399** that reference solve itself stopped at its iteration limit after 1770 EVDs with $\|\nabla\theta\|_2 = 4.59 \times 10^{-13}$. At $n \approx 3000$ an absolute tolerance of 10^{-13} is not attainable in double precision, so for **cor3120** and **bccd16** the reference solve is capped at 30 EVDs and its achieved residual is reported instead: $\|\nabla\theta\|_2 = 1.58 \times 10^{-12}$ and 8.07×10^{-13} , after 2201s and 895s. A second, independent reference for each (SBB-Dual driven to its floor, then the same Newton polish) agrees with the first to $\|X_A - X_B\|_F = 1.3 \times 10^{-11}$ and 8.1×10^{-12} , so forward errors on these matrices are resolved well below the targets used; the residual floor near 10^{-12} is reached within four to six Newton iterations. We therefore treat these runs as a check on external validity, not as measurements of the same quality as table 5. Every solver is run to a common tolerance of 10^{-9} , capped at 500 EVDs on the two smaller matrices and at 300 iterations on the two larger ones, and costs are read at the forward-error target 10^{-8} .

The matrices disagree with each other, broadly in the direction the synthetic family predicts. At $n = 94$, SBB-Dual needs 7 EVDs to 10^{-8} against 12 for AGD-SDAJ-BH and 11 for Dykstra-APM, which is the pattern of table 6 at $n = 100$, and Anderson-APM needs 6. At $n = 1399$,

Table 13: Invalid correlation matrices from the Anymatrix **CORRINV** collection [17], run to a common tolerance of 10^{-9} . EVDs to reach and hold forward error 10^{-8} , total EVDs at the solver’s own exit, and the forward error there, all measured against a computed reference solution, not an exact one. [‡]Reaches its cap and is censored. Five solvers come from one run; Anderson-APM was run later against a recomputed reference, and a replay of AGD-SDAJ-BH against that reference reproduces every cost to target in this table.

matrix	solver	EVDs to 10^{-8}	total EVDS	final error
Rocky Mountain $n = 94$	Newton-SIN-BH	4	4	1.49×10^{-13}
	Anderson-APM	6	7	2.52×10^{-10}
	SBB-Dual	7	8	1.53×10^{-10}
	Dykstra-APM	11	13	2.57×10^{-10}
	AGD-SDAJ-BH	12	17	3.47×10^{-10}
	AGD-SDAJ	12	25	3.47×10^{-10}
cor1399 $n = 1399$	Newton-SIN-BH	7	7	1.27×10^{-9}
	AGD-SDAJ-BH	66	69	1.98×10^{-9}
	AGD-SDAJ	67	70	1.90×10^{-9}
	Anderson-APM	103	109	3.08×10^{-9}
	SBB-Dual	200	212	3.01×10^{-9}
	Dykstra-APM	406	430	3.17×10^{-9}
cor3120 $n = 3120$	Newton-SIN-BH	6	6	2.33×10^{-9}
	AGD-SDAJ-BH	41	42	1.32×10^{-9}
	AGD-SDAJ	56	59	2.17×10^{-10}
	Anderson-APM	155	160	3.21×10^{-9}
	SBB-Dual	255	270	3.38×10^{-9}
	Dykstra-APM [‡]	–	300	1.16×10^{-5}
bccd16 $n = 3250$	Newton-SIN-BH	4	5	2.71×10^{-11}
	SBB-Dual	5	6	1.20×10^{-10}
	Anderson-APM	6	6	4.22×10^{-10}
	Dykstra-APM	6	7	1.46×10^{-10}
	AGD-SDAJ-BH	8	9	8.88×10^{-10}
	AGD-SDAJ [‡]	–	323	3.93×10^{-5}

SBB-Dual costs three times AGD-SDAJ-BH (200 against 66), and at $n = 3120$ six times (255 against 41), which is the pattern of section 5.6. **bccd16** is easy for every method: all but one reach 10^{-8} within 8 EVDS, so it does not discriminate between the first-order methods. Newton-SIN-BH is cheapest on all four, as everywhere else in this paper. The change of order is therefore visible on matrices with uncontrolled spectra that we did not construct. With four matrices that differ in more than dimension, this is consistent with the dimension effect of section 5.6, not an independent demonstration of it.

The line-search stall occurs on real data. On the Rocky Mountain matrix the naive-Armijo Newton control exhausts its budget at a forward error of 1.03×10^{-8} , while the same solver with the Borsdorf–Higham line search converges in 4 EVDS to 1.49×10^{-13} . On **cor1399** the naive control converges normally (13 EVDS against 7). On **bccd16** the same defect appears in unmodified AGD-SDAJ. Both AGD variants follow an identical path to a forward error of 3.93×10^{-5} at the fifth EVD. From there AGD-SDAJ-BH, whose safeguard accepts the step, converges in four more EVDS to 8.9×10^{-10} ; unmodified AGD-SDAJ rejects trial after trial in its monotone line search, with the

error unchanged in its first eight digits, until it exhausts the budget at 323 EVDS. The defect of section 4.2 is thus not an artifact of the constructed family, and it affects both published methods on real data, but neither is it universal: whether it appears depends on the matrix. Unmodified AGD-SDAJ also stalls on 10 of the 81 data-seeded and 1 of the 20 perturbed instances of section 5.8, and on mmb13, tyda99r2 and bhwi01 (section B).

The cost of a reference. At $n \approx 3000$ the reference solve alone took 15 to 37 minutes, and a first attempt at bccd16 with an uncapped absolute tolerance ran for four hours without finishing. A reference accurate enough to measure forward error can cost more than the solves being measured. That is a practical argument for the synthetic family, whose solution is available in closed form at any size.

5.11 Accounting convention

Under all-trial accounting, every evaluated point counts toward reaching and holding a target, including rejected line-search trials. A rejected trial can then be credited with reaching a target, but one that lies above the target also breaks a hold. On this family the second effect dominates slightly. Median costs change only for the two AGD variants, and all-trial accounting is marginally the more expensive: for AGD-SDAJ-BH the median rises by 1.0, 1.5 and 2.0 EVDSat $\varepsilon = 10^{-4}, 10^{-8}, 10^{-10}$, and for unmodified AGD-SDAJ by 1.0, 2.0 and 2.0. Newton-SIN-BH, SBB-Dual and Dykstra-APM are unaffected, because they have few or no rejected trials here. The effect is negligible on this family and never changes an ordering. For methods with many rejected trials it need not be, and accepted-only accounting is the rule that cannot credit a point the method itself rejected.

Two further accounting choices deserve a check, because each could favour a particular method. First, cost is scored on the raw projection $\Pi_{\mathcal{S}_+^n}(G + \text{Diag}(y))$, while SBB-Dual applies a Borsdorf–Higham rescaling before returning its iterate (section 5.1), so it is scored on a quantity it does not return. Both columns are recorded. Rescoring every study on the rescaled iterate leaves the ordering unchanged at 19 of the 20 (dataset, target) pairs, and moves SBB-Dual’s median at $n = 500$ and $\varepsilon = 10^{-8}$ from 207.5 to 188 EVDS: the reported numbers are the conservative choice for our own method. The single exception is the data-seeded set at 10^{-10} , where rescoring lowers SBB-Dual’s reach count from 81 to 45 instances, so the apparent change of order is a comparison over an easier subset rather than a reversal. Second, the AGD line searches return their last trial if a backtracking limit of 60 is reached, and the method then moves to a point the trajectory records as rejected. This never occurs in the reported runs: no line search in any study exceeded 60 trials, and every AGD stall recorded here is a work-budget exhaustion, not a backtracking failure.

The iteration caps are not uniform across methods, and no comparison in this paper relies on a capped run. At $n = 500$ the caps are 1200 iterations for SBB-Dual, 800 for Dykstra-APM and a 4000-EVD budget for the AGD variants, chosen so that each method’s own exit is reached on most instances within a fixed total runtime. SBB-Dual converged on all 270 instances (worst case 1075), so its numbers are uncensored, and the AGD-SDAJ medians are taken over the instances that reached each target, which understates the cost of the unmodified variant and therefore understates the benefit of the safeguard.

Dykstra-APM is the one method whose cap binds, on 90 of 270 instances, and it also had the smallest budget, so we re-ran those 90 at SBB-Dual’s cap of 1200 iterations. *All 90 reached the larger cap as well.* At 1200 iterations they end at a median forward error of about 10^{-6} : 71 of the 90 reach 10^{-6} (median 1099 EVDS, against 152 for SBB-Dual and 53.5 for AGD-SDAJ-BH at the same target), 6 reach 10^{-8} , and none reaches 10^{-10} . The censoring is therefore a property of the method on these instances and not of the budget it was given; a cap generous enough to uncensor

Table 14: Feasibility of the returned iterate at matched tolerance 10^{-11} , 270 instances. ‘diag err’ is $\|\text{diag}(X) - e\|_\infty$.

solver	median $\lambda_{\min}(X)$	worst $\lambda_{\min}$	median diag err	worst diag err
Newton-SIN-BH	-3.42×10^{-15}	-1.25×10^{-14}	8.38×10^{-15}	5.70×10^{-12}
AGD-SDAJ-BH	-3.40×10^{-15}	-1.36×10^{-14}	2.53×10^{-12}	7.20×10^{-12}
AGD-SDAJ	-3.40×10^{-15}	-1.39×10^{-14}	2.79×10^{-12}	4.01×10^{-8}
SBB-Dual	-3.42×10^{-15}	-1.49×10^{-14}	0	0
Dykstra-APM	-3.43×10^{-15}	-1.36×10^{-14}	1.49×10^{-12}	9.37×10^{-12}
Anderson-APM	-3.45×10^{-15}	-1.36×10^{-14}	2.12×10^{-12}	7.50×10^{-12}

Dykstra-APM would have to be an order of magnitude larger than any other method needs. We report its position as censored throughout.

5.12 What each solver returns

Methods also differ in what they return (table 14). All six return matrices with $\lambda_{\min}$ at roundoff level, so none is meaningfully more positive semidefinite than another. The diagonal is a different matter. SBB-Dual returns an exactly unit diagonal because of its rescaling epilogue, whereas every other variant returns a diagonal that is accurate only to its convergence tolerance. The worst-case diagonal error of unmodified AGD-SDAJ is 4.01×10^{-8} , about 3.6 orders of magnitude larger than that of any other variant, which is a direct consequence of the stalls in section 4.2. The rescaling epilogue has a cost of its own: on a nearly feasible input at loose tolerance, it can increase $\|X - X^*\|_F$. Borsdorf and Higham bound that increase by $\text{tol} \|X\|_F / (1 - \text{tol})$ [6, Lem. 3.2], which is small whenever the tolerance is tight but need not be at a loose one. A benchmark that reports only cost hides these differences. A user who needs an exactly unit diagonal and a user who needs minimal distance may not be best served by the same method.

Costs at each solver’s own exit when all are run to 10^{-11} , and the exit reasons behind them, are in the released analysis output; under that convention the ordering equals the tight-target ordering of section 5.3, because the common tolerance is tight.

6 The dimension-scaled stopping rule

A common exit criterion for dual NCM methods is $\|\nabla\theta(y_k)\|_2 \leq 10^{-7}n$. It is used, for example, in the experiments of [19]. Because the threshold grows linearly with n , the forward error it guarantees depends on both the dimension and the method. The general point that native stopping tests confound comparisons was made by Dolan, Moré and Munson [11]. This section documents a specific NCM instance of it.

Spread across methods at a fixed n . Table 15 and fig. 5 report the timing run under this rule at $n = 100$, where the tolerance is 10^{-5} .

At this dimension the best and worst median achieved accuracies differ by a factor of 383 over all 270 instances. Repeating the run at $n = 500$ on a subset of 18 instances (threshold 5×10^{-5}) gives a spread of only 13.8, with medians from 1.74×10^{-5} for Newton-SIN-BH to 2.41×10^{-4} for Dykstra-APM. The spread across solvers is therefore not itself a stable quantity, and we do not claim it as one; the next paragraph compares the two dimensions on the same instances. Newton overshoots the tolerance because a quadratically convergent step moves far past the threshold.

Table 15: Solvers exiting when the stopping quantity specific to each method first reaches the common numerical threshold $10^{-7}n$ at $n = 100$ (for the dual methods this quantity is $\|\nabla\theta\|_2$; for Dykstra-APM and Anderson-APM it is the diagonal error of the PSD iterate), 270 instances, single-threaded timing.

solver	median EVDs	median time (s)	median $\ X - X^*\ _F$
Newton-SIN-BH	4	0.006	5.04×10^{-8}
Newton-SIN	7	0.009	5.04×10^{-8}
AGD-SDAJ-BH	20	0.024	9.58×10^{-6}
AGD-SDAJ	20	0.024	9.58×10^{-6}
SBB-Dual	18	0.020	7.73×10^{-6}
Dykstra-APM	40	0.045	1.93×10^{-5}
Anderson-APM	13	0.016	1.37×10^{-5}

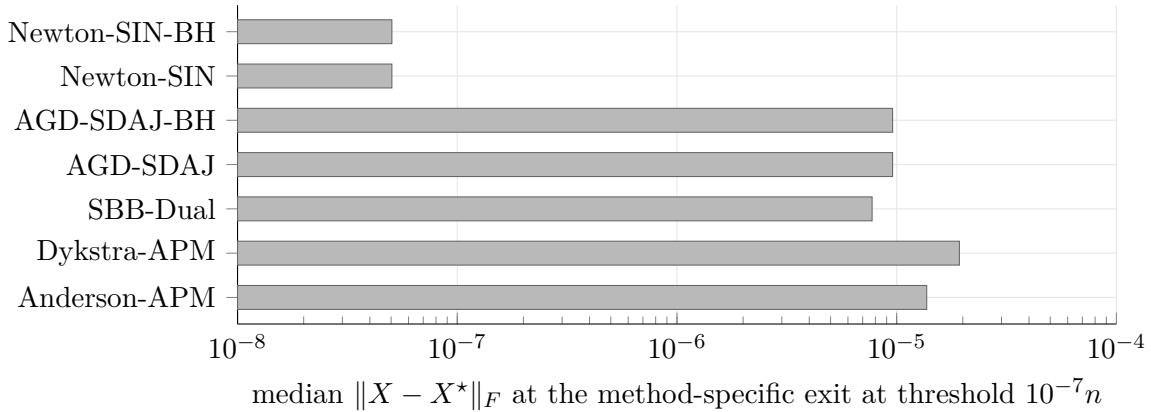


Figure 5: Achieved accuracy when each solver exits under the same published dimension-scaled rule (tolerance 10^{-5} at $n = 100$). Median forward errors range from 5.04×10^{-8} (Newton variants) to 1.93×10^{-5} (Dykstra-APM), a factor of 383. Costs measured at these exits are therefore costs at different accuracies.

First-order methods stop close to it, and Dykstra-APM’s stopping quantity (the diagonal error of its PSD iterate) is not the dual gradient norm at all. The cost column is therefore a comparison at different accuracies, and it favours methods that stop early. At this tolerance the native costs also place SBB-Dual (18 EVDs) slightly ahead of both AGD variants (20 EVDs), with Anderson-APM ahead of all three (13 EVDs). That comparison sits at the crossover of fig. 2: the accuracies reached, near 10^{-5} , lie where the matched-accuracy order changes, so the native ranking agrees with neither the loose-target nor the tight-target regime and should not be read as evidence for either.

Growth with n . The rule loosens as n grows. In earlier runs, outside the ranking study, a nearly feasible $n = 500$ instance exited after *zero* updates with forward error 7.8×10^{-4} . At $n = 2000$, both dual first-order methods stopped after 3 to 5 iterations with forward errors between 2.7×10^{-3} and 4.1×10^{-3} . Under this rule, a larger problem can appear easier simply because less accuracy is being asked for. The paired family makes the same point with the dimension controlled. Each of the 18 instances run at $n = 500$ has a counterpart at $n = 100$ with the same rank, multiplicity, separation and seed label, so the comparison can be made instance by instance. The threshold loosens by a factor of five between the two. The first-order methods degrade steadily: the paired

median ratio of the forward error at $n = 500$ to that at $n = 100$ is 10.1 for AGD-SDAJ-BH, 7.3 for SBB-Dual, 12.6 for Dykstra-APM and 9.9 for Anderson-APM. Per instance these ratios lie between 4 and 25 for the first three and between 3.8 and 42 for Anderson-APM. Newton-SIN-BH degrades by a paired median factor of 1.6×10^3 , but erratically, from 0.16 to 5.3×10^5 across the 18 instances. Its exit lands far inside the threshold, at a median gradient norm 8×10^{-4} times the threshold at $n = 100$ but 0.20 times it at $n = 500$: quadratic convergence moves in large discrete steps, so the margin by which it overshoots is itself erratic and collapses whenever the run exits one step earlier. The first-order methods stop at 0.45 to 0.99 of the threshold at both dimensions and so track it. On these 18 instances the spread across solvers therefore narrows from 1653-fold at $n = 100$ to 13.8-fold at $n = 500$: the rule does not degrade the methods equally, and it conceals most of the advantage of the method that converges fastest. A reader comparing published $n = 100$ and $n = 500$ results under this rule is comparing solutions of very different quality. The real-matrix runs of section 5.10 were made at a matched tolerance only, so they give no native-rule spread on real data.

Recommendation. Timing or iteration counts should never be reported without the achieved forward error (or, when X^* is unknown, a certified bound on it). Where a dimension-scaled rule is used for compatibility with earlier work, it should be accompanied by a run to a common tight tolerance.

7 Reproducibility

Independent implementations of the earlier solvers. SBB-Dual, semismooth Newton and Dykstra-APM were implemented independently in Python and in Julia, using *different* constructions of the reference solution. On all 14 shared instances, the two implementations agree on update counts and forward errors to every printed digit.

AGD-SDAJ: verified against the published counts. AGD-SDAJ exists only in Julia. Because this paper argues that implementation details can reverse conclusions, we did not report any result for our reimplementations on the KKT family until it had reproduced Huynh and Hwang’s own published counts on their own instances. Those instances are three matrices from their test set, loaded under the same convention used elsewhere in this project (table 16).

P9 and P8 agree exactly. P7 agrees on outer iterations and residual norm, and differs by one backtracking decision: 7 rejected trials locally against 8 published. That single decision accounts for the whole difference in EVDs. Excluding backtracks, the local base cost is $32 - 7 = 25$ and the published base cost is $31 - 8 = 23$. The gap of 2 EVDs is exactly the early-termination saving described in the note to their Table 3: with the extra backtrack, their trajectory reaches the tolerance *at* the preconditioned point and skips the final gradient step, whereas ours reaches it one step later. We attribute the differing decision to floating-point differences. The reference implementation is in MATLAB and ours in Julia. The two call different LAPACK eigensolver drivers, and on a dense 1399×1399 problem, differences in the last bits of the spectrum are enough to flip a single acceptance test whose two sides are nearly equal. We did not search over parameters to make P7 match, because tuning to a target would be exactly the kind of hidden fitting this paper warns against. Section B gives the full verification record, including three transcription defects found and corrected along the way.

Single-machine caveat. All timings come from one machine with single-threaded BLAS: Intel Core 7 150U (10 physical cores, 12 logical), 32GB RAM, Windows 11 (build 26200), Julia 1.12.6

Table 16: AGD-SDAJ ($q = 2$): published counts (outer iterations / EVDs / nonmonotone backtracks / $\|A - X^*\|_F$) against our reimplementaion. Published values are from Table 3 and Table A1 of [19].

instance	published	this work	status
P9 bccd16, $n = 3250$	1 / 5 / 0 / 29.06	1 / 5 / 0 / 29.0563	exact
P8 cor3120, $n = 3120$	3 / 18 / 1 / 5.44	3 / 18 / 1 / 5.4437	exact
P7 cor1399, $n = 1399$	4 / 31 / 8 / 21.03	4 / 32 / 7 / 21.0339	one backtrack differs

with OpenBLAS (ILP64) through libblastrampoline. Per-EVD ratios (table 3) may differ on other hardware and BLAS libraries, and with multithreading. Each configuration was timed once, after a discarded warm-up, so we do not report run-to-run variance.

Artifact. The generator, the instances with hashes, all solver code, the trajectory and timing CSV files, and the analysis scripts that produce every table in this paper are listed in section D. They are available at <https://github.com/Realosunboy6/ncm-exact-benchmark>. An archived snapshot carrying a permanent identifier will be deposited on acceptance.

8 Discussion and limitations

What changed the conclusions, and what did not. On this family, none of the conventions we tested changed which method was cheapest. The accuracy target, the instance regime and the problem dimension did change the order below it: the changes of lower-place ordering with the accuracy target and with rank that we found at $n = 100$ recur at $n = 500$ only above $r = 50$ (sections 5.6 and 5.7), and on matrices built from equity returns the degree of invalidity reverses the order as well (section 5.8). The stopping rule changed the achieved accuracy by a factor of 383. The cost unit (EVDs against wall-clock time) and the accounting rule (accepted-only against all-trial) changed cost ratios but no ordering. Two line-search details changed whether methods converged at all. We think the correct summary is conservative. The protocol matters mainly for the *magnitude* of reported advantages and for the ordering of methods that are close in cost. On the evidence here, it does not matter for identifying a clear first place. We expect this to hold wherever one method is ahead by a large factor; where methods are within a factor of about two, conclusions drawn under a single convention should be treated with caution.

Why an exact family. The regime dependence in section 5.5 is organized mainly by rank, one of the three quantities the construction controls. Finding it requires both an exact X^* , so that forward error is not contaminated by an approximate reference, and independent control of the spectral structure at the solution. Real matrices supply neither. The same machinery then showed the dimension dependence of that finding: at $n = 500$ the same rank effect changes the order only between $r = 50$ and $r = 75$ (section 5.7), so a grid chosen at one dimension can miss the crossover at another.

Limitations.

- *Problem size.* Results are reported at $n = 100$ and $n = 500$ (270 instances each), with 135 more at $n = 500$ and $r \in \{75, 100, 150\}$ for every variant except unmodified AGD-SDAJ. The crossover between SBB-Dual and AGD-SDAJ-BH lies between $r = 5$ and $r = 20$ at $n = 100$ and between $r = 50$ and $r = 75$ at $n = 500$; on data-seeded instances at $n = 550$ SBB-Dual

is still slightly behind at $r = 100$, and at $n = 2105$ (section 5.9) the gap closes to within one EVD by $r = 400$ without a resolved crossover. The crossover is therefore bracketed at each dimension and rises with it; the data do not support a functional form, and the largest panel contributes four instances per rank with weak replication. Dykstra-APM is censored at $n = 500$ by its iteration cap.

- *Synthetic family.* The KKT family is synthetic. Its G matrices are correlation-like but are not estimated from data. Section 5.10 adds four matrices from the literature, from $n = 94$ to $n = 3250$, measured against a computed rather than an exact reference; for the two largest that reference is capped and reaches $\|\nabla\theta\|_2 \approx 10^{-12}$ rather than 10^{-13} , and no wall-clock times were taken. Four matrices cannot separate dimension from the other ways in which they differ. The equity-based instances of section 5.8 use a factor model of one correlation matrix as the exact solution; those instances are only mildly invalid and have three replicates per setting, while the perturbed set has five draws per σ and a computed reference. The sets of section 5.8 rest on one base correlation matrix and those of section 5.9 on a second, so there are two panels in all, not a sample of them.
- *Statistical uncertainty.* Aggregate medians at both dimensions carry cluster-bootstrap intervals (tables 5 and 7). Each dimension rests on five paired seeds per rank, that is 15 seed clusters, so inferential statements remain modest. Cell-level orderings rest on five instances each, and within a cell the adjacent pair AGD-SDAJ-BH against AGD-SDAJ is usually unresolved, so the count of distinct orderings should not be read as a count of real regime changes. We report counts and intervals rather than significance tests, because the reuse of each drawn U across all cells of its rank makes instance-level comparisons strongly dependent.
- *Single machine.* All timings come from one machine with single-threaded BLAS.
- *Our adaptation.* The Borsdorf–Higham adaptation to AGD-SDAJ is ours, not the original authors’. It is reported beside the unmodified method throughout.
- *Feasibility propositions.* Proposition 2.3 assumes uniform μ , and the extension to mixed μ is open (section A). Proposition 2.5 is stated for a only; no companion bound on b is given, since the released family lies in the fixed-rank regime where proposition 2.5 does not apply.
- *Scope of methods.* We benchmark six variants of five methods. Not included are the recent semismooth Newton method for general projection equations [2] and the factor-structured variant [7], and Anderson acceleration is run only at the authors’ default history length $m = 2$.
- *Attribution of the line search.* We use the Borsdorf–Higham globalization but not the rest of their algorithm, and the adaptation of their near-equality safeguard to AGD-SDAJ is ours (sections 4.2 and 5.1).
- *Unweighted problem.* Only the unweighted Frobenius problem is considered.

Conclusion. An exact instance family with a prescribed solution structure makes it possible to separate stopping-rule, reference, accounting and primitive-work effects in NCM benchmarks. On 270 instances at $n = 100$, these effects changed cost ratios and the lower places of the ranking, but not the cheapest method. Repeating the study on 270 instances at $n = 500$ sharpened that finding: Newton-SIN-BH remains the leading method, and the changes of lower-place ordering with

the accuracy target and with rank recur, but only above $r = 50$, a rank the original grid did not reach. Rankings below the leading method are therefore not intrinsic properties of the solvers; they depend materially on the benchmark regime, including its dimension, the rank of the solution and the size of the violation. The one clear failure we found is a known finite-precision line-search defect, which appears in two separately published methods, on the constructed family and on real matrices. It is removed in the semismooth Newton method by the published Borsdorf–Higham modified line search, and in AGD-SDAJ by our adaptation of their near-equality safeguard. We recommend that NCM comparisons report forward error against a trusted solution at a common tolerance, give rejected trials no accuracy credit, and report primitive work and wall-clock time together.

A Proofs for the instance family

Throughout, $U \in \mathbb{R}^{n \times r}$ has rows u_i , $X^* = UU^\top$, P_r is the orthogonal projector onto $\text{range } U = \text{range } X^*$, and $Q_0 Q_0^\top = I - P_r$ (because X^* is symmetric, $\ker X^*$ is the orthogonal complement of $\text{range } X^*$).

A.1 Proof of proposition 2.3

With $\mu = ce$, we have $W = cQ_0 Q_0^\top = c(I - P_r)$. For $i \neq j$, the diagonal correction $\text{Diag}(\text{diag } W)$ in (3) does not affect the entry (i, j) , and $I_{ij} = 0$, so

$$G_{ij} = X_{ij}^* - W_{ij} = X_{ij}^* + c(P_r)_{ij}.$$

By the triangle inequality, $|G_{ij}| \leq |X_{ij}^*| + c|(P_r)_{ij}| \leq a + cb$. If $a < 1$ and $c \leq (1 - a)/b$, then $a + cb \leq 1$. $\square$

Remark A.1. The bound is sufficient but not necessary. Cancellation between X_{ij}^* and $c(P_r)_{ij}$, and the fact that a and b are generally attained at different pairs (i, j) , both make it conservative. For this reason the generator checks $|G_{ij}| \leq 1$ entry by entry.

A.2 Proof of proposition 2.4

Fix $r \geq 2$ and $\epsilon \in (0, 1)$. Because S^{r-1} is compact, it can be covered by finitely many caps $C_1, \dots, C_N$, $N = N(\epsilon)$, each of angular radius $\epsilon/2$. Any two points in the same cap make an angle of at most ϵ , so their inner product is at least $\cos \epsilon \geq 1 - \epsilon^2/2$. Once $n > N$, the pigeonhole principle forces two of $u_1, \dots, u_n$ into a common cap, so $a \geq 1 - \epsilon^2/2$ deterministically. Since ϵ is arbitrary and $a \leq 1$ always, $a \rightarrow 1$ along every realization. (A probabilistic argument is needed only for a *rate*; the limit itself follows from pigeonhole.) $\square$

Remark A.2 (What this does not give). It is tempting to conclude that a fixed $c > 0$ must eventually violate proposition 2.3. That does not follow. At fixed r the projector bound b also tends to zero, so $a + cb \leq 1$ can persist: with $a_n = 1 - 2/n$ and $b_n = 1/n$ one has $a_n \rightarrow 1$ while $a_n + cb_n < 1$ for every n at $c = 1$. A claim in either direction needs the relative rates of $1 - a$ and b , which we do not establish here.

A.3 Proof of proposition 2.5

Fix $i \neq j$ and condition on u_i . Since u_j is uniform on S^{r-1} and independent of u_i , we use the classical spherical-cap bound: for every unit vector v and every $t \geq 0$,

$$\Pr\{u_j^\top v \geq t\} \leq 2 \exp(-t^2 r/2).$$

This explicit form is the cap estimate of [3, Lem. 2.2]; the subgaussian statement of [27, Thm. 3.4.5] gives the same decay with an unspecified absolute constant in the exponent, which would leave the constant in proposition 2.5 unspecified as well. Applying this with $v = u_i$, and again to $-u_j$, gives $\Pr\{|u_i^\top u_j| \geq t\} \leq 4 \exp(-t^2 r/2)$; the bound is unchanged after averaging over u_i . Taking a union bound over the fewer than $n^2/2$ unordered pairs,

$$\Pr\{a \geq t\} \leq 2n^2 \exp(-t^2 r/2).$$

Setting the right-hand side equal to η and solving for t gives $t = \sqrt{(2/r)(2 \log n + \log(2/\eta))}$, which is the stated bound. If $r/\log n \rightarrow \infty$ then $t \rightarrow 0$ for any fixed η . $\square$

Remark A.3 (On b). We do not state a companion bound for $b = \max_{i \neq j} |(P_r)_{ij}|$. Writing $P_r = VV^\top$ with V having orthonormal columns, $(P_r)_{ij}$ is an inner product of two rows of V , whose joint distribution is not that of two independent uniform vectors; a bound would need a model for range U (for instance a Haar-random subspace) together with a justification that the row-normalized Gaussian U induces it. Since the released family lies in the fixed-rank regime of proposition 2.4, where proposition 2.5 does not apply, and since the generator screens $|G_{ij}| \leq 1$ entrywise in any case, we leave this open.

Remark A.4 (Scope). These propositions concern $|G_{ij}| \leq 1$ only. Indefiniteness of G ($\lambda_{\min}(G) < 0$) is not implied and is checked separately. The released family ($r \in \{5, 20, 50\}$, $n = 100$) is in the fixed-rank regime of proposition 2.4, not the high-dimensional regime of proposition 2.5.

A.4 Mixed μ

For the released $\mu = (\mu_{\text{bulk}} e_{k-m}, \delta e_m)$, write $Q_0 = [Q_b \ Q_\delta]$. Then $W = \mu_{\text{bulk}} Q_b Q_b^\top + \delta Q_\delta Q_\delta^\top = \mu_{\text{bulk}}(I - P_r) - (\mu_{\text{bulk}} - \delta) Q_\delta Q_\delta^\top$. The off-diagonal entries therefore pick up an extra term $(\mu_{\text{bulk}} - \delta)(Q_\delta Q_\delta^\top)_{ij}$, whose size depends on the choice of basis for $\ker X^*$ used to select Q_δ . A bound analogous to proposition 2.3 follows with b replaced by $b + \max_{i \neq j} |(Q_\delta Q_\delta^\top)_{ij}|$.

Remark A.5 (Open). We do not characterize $\max_{i \neq j} |(Q_\delta Q_\delta^\top)_{ij}|$. It depends on which orthonormal basis of $\ker X^*$ is chosen, and the generator fixes that basis through the eigendecomposition it computes, so the quantity is not determined by (n, r, m, δ) alone. A basis-free bound would need either a canonical choice of Q_δ or a bound uniform over admissible bases. We therefore state proposition 2.3 for uniform μ only, and the generator screens the mixed- μ instances entrywise rather than relying on a bound. This is why the released family is screened rather than certified.

B AGD-SDAJ verification record

Gate. Before running our reimplementaion of AGD-SDAJ [19] on the KKT family, we required it to reproduce the authors' own published counts on their own instances. An unverified reimplementaion that is then reported as losing would not be a defensible result.

Source and implementation. The source is Huynh and Hwang (2025), *Numer. Linear Algebra Appl.* 32(6), DOI 10.1002/nla.70045. We transcribed Algorithm 1 (AGD), Algorithm 3 (QN-SDAJ), Algorithm 4 (AGD-SDAJ) and the Section 4.3 parameters from the full text: $c = 10^{-4}$, $\rho = 0.5$, $\lambda_1 = \lambda_2 = 0.01$, $\epsilon_l = 10^{-5}$, $\epsilon_u = 10^8$, $q = 2$, $x_0 = 0$, the stopping rule $\|\nabla \theta\| \leq 10^{-7}n$, and $k_{\max} = 200$. The implementation is `agd_sdaj.jl` (standalone, used for verification) and `agd_sdaj_ncm` in `bench_sbb_dual.jl` (instrumented, used for the study). The instances are the Anymatrix CORRINV files already used elsewhere in this project, exported with the same loading convention.

Equation (6), as transcribed. The nonmonotone line search of Algorithm 3 is a Li–Fukushima-type rule [22] on the residual:

$$\|\nabla\theta(x + \alpha d)\|^2 \leq (1 + \epsilon_k)\|\nabla\theta(x)\|^2 - \lambda_1\|\alpha\nabla\theta(x)\|^2 - \lambda_2\|\alpha d\|^2, \quad \epsilon_k = \frac{1}{(k+1)^2}, \quad (6)$$

with $\lambda_1 = \lambda_2 = 0.01$ and $\alpha = \rho^j$ backtracked from 1 over $j = 0, 1, 2, \dots$ (we write j for the backtracking exponent, since m is the near-zero multiplicity of proposition 2.1). The merit function is $\frac{1}{2}\|\nabla\theta\|^2$. The quasi-Newton direction $d = -B^{-1}\nabla\theta$ need not be a descent direction for it, which is why a nonmonotone rule is needed. The paper also confirms that B_k is reset to the identity once x_k^{pre} is generated, and that Step 4 discards x_k^{pre} if it does not decrease θ .

Before the full text was available, we tried two guesses: a Grippo–Lampariello–Lucidi rule on θ , and a Li–Fukushima rule with unsquared norms and $\epsilon_k = 0$. Both had the right general form but three wrong details: unsquared norms, a missing $(1 + \epsilon_k)$ slack, and λ_1, λ_2 attached to the wrong terms. Because the two guesses bracketed the published counts, they helped locate the remaining gap.

Table 17: Outer iterations / EVDS / nonmonotone backtracks, and $\|A - X^*\|_F$, under the final transcription (eq6) and the two superseded guesses.

instance	published	eq6 (final)	GLL guess	LF guess
P9 bccd16	1/5/0, 29.06	1/5/0, 29.0563 (exact)	1/5/0	1/5/0
P8 cor3120	3/18/1, 5.44	3/18/1, 5.4437 (exact)	3/17/0	3/19/2
P7 cor1399	4/31/8, 21.03	4/32/7, 21.0339	6/40/3	5/48/17

Results. Under every variant, $\|A - X^*\|_F$ matched the published value to the displayed precision on all three instances. This showed early on that the objective, gradient, projection and stopping rule were correct, and that the discrepancy lay in the amount of work per iteration.

Three hazards when reimplementing Algorithm 4. Each of the following is easy to introduce when transcribing the method from its pseudocode. Each leaves the method converging to the correct solution while changing the work it does, so none is visible from final errors, only from work counts checked against the published ones. We record them for anyone reimplementing the method. All three occurred in our own first transcription and were found by exactly that check: the first two by accounting for the five published EVDS on P9 one at a time, the third because a gap in outer iterations on P7 remained once the line search was correct.

1. *Redundant EVD at an already-evaluated point.* The inner QN-SDAJ routine recomputed θ and $\nabla\theta$ at the point its caller had just evaluated. This is the same class of defect as section 4.1, and it was found in the same way. Fix: pass the caller’s values into the inner routine.
2. *Missing early termination at the preconditioned point.* The note to Table 3 of [19] states that the loop also stops whenever x_k^{pre} satisfies the stopping criterion, which saves fewer than two EVDS. Without this check, the outer gradient step is always paid.
3. *Wrong acceleration factor in Algorithm 3.* The factor β_k^{pre} had been implemented with Algorithm 1’s $a_k = \|g\|^2$. That formula is the special case for $d = -\nabla\theta$. In Algorithm 3, $d = -B^{-1}g$, so the general form of their Section 2.3 (the acceleration of Andrei [1]) is needed:

$$\beta = \frac{-g^\top d}{(g_y - g)^\top d} \quad \text{if the denominator is positive, else } \beta = 1.$$

This form reduces exactly to Algorithm 1’s when $d = -g$, so it is consistent with both statements in the paper. *This defect caused the gap in outer iterations*: correcting it reduced P7 from 6 outer iterations to 4.

The cumulative effect on P7 was 46 EVDS $\rightarrow$ 40 (defects 1–2) $\rightarrow$ 43 (exact equation (6)) $\rightarrow$ 32 (correct acceleration factor), against a published 31.

The remaining P7 residual. See section 7. P7 matches in outer iterations (4) and residual norm (21.0339) and differs by one backtracking decision (7 against 8). That decision accounts for the whole difference in EVDS, through the early-termination saving. We attribute it to last-bit differences between the MATLAB and Julia eigensolvers. P8 and P9 reproduce every count exactly on the same code path. No further parameter search was carried out.

Conditions of use (all met in this paper). (i) The code and this record are published with the results. (ii) The P7 residual is disclosed. (iii) Published counts are printed beside local counts wherever the comparison appears (tables 16 and 17).

Checks against other published results. Beyond the AGD-SDAJ gate, every solver was run on invalid correlation matrices with published solutions or work counts (table 18). The matrices are those of the CORRINV collection [17], taken from its repository rather than retyped. On the ten small matrices (`high02`, `tec03`, `bhwi01`, `mmb13`, `fing97`, `beyu11`, `tyda99r1–r3`, `usgs13`), run at tolerance 10^{-10} , the five solvers of the original comparison return valid correlation matrices (Anderson-APM, added later, is checked in section C) whose distances $\|G - X\|_F$ agree to at least eight digits and whose matrices agree with Newton-SIN-BH’s to within 2×10^{-10} , except unmodified AGD-SDAJ on `mmb13` (8.7×10^{-10}), which stops at its budget. The three large matrices were compared only at the published tolerances.

Table 18: Work counts reproduced from the literature. Dykstra-APM uses the stopping rule of the released code `nearcorr_new.m` of [18], tolerance n eps; run for the same number of iterations, our implementation returns the same iterate, bit for bit on nine matrices and to 1.4×10^{-15} on `mmb13`. Newton-SIN-BH is compared with Tables 6.5 and 6.6 of [5].

solver	source	matrix	published / ours
Dykstra-APM	[18], Table 1	<code>tec03</code> , $n = 4$	39 / 39 iterations
		<code>bhwi01</code> , $n = 5$	27 / 27
		<code>mmb13</code> , $n = 6$	801 / 801
		<code>fing97</code> , $n = 7$	33 / 34 ^a
	[18], Table 4	<code>usgs13</code> , $n = 94$	18 / 18
Newton-SIN-BH	[5]	<code>cor1399</code> , $10^{-7}n$	5 / 5 iterations
		<code>cor1399</code> , n eps	7 / 7; $\ G - X\ _F = 21.033911$ / 21.033911
		<code>cor3120</code> , $10^{-7}n$	4 / 4
		<code>cor3120</code> , n eps	6 / not reached ^b ; 5.443751 / 5.443751

^aAt iteration 33 the stopping quantity is 1.12 times the tolerance 7 eps, so the extra iteration is decided in the last bits. ^bThe residual floor near 10^{-12} lies above n eps = 6.9×10^{-13} with our eigensolver; the solution itself matches to all published digits.

On three of these matrices unmodified AGD-SDAJ shows the stall of section 4.2 (`mmb13`: budget of 200 000 EVDS exhausted, against 210 for AGD-SDAJ-BH; `tyda99r2`: 549 against 35; `bhwi01`: 66 against 21).

The Borsdorf–Higham adaptation. AGD-SDAJ-BH changes only the monotone Armijo test of Algorithm 1, line 5 (algorithm 2). When $|\theta_t - \theta| < \gamma u(1 + |\theta_t| + |\theta|)$ with $\gamma = 100$ and $u = 2^{-53}$, a trial that reduces $\|\nabla\theta\|$ is accepted. The nonmonotone rule (6) is unchanged. This adaptation is ours.

C Anderson-APM verification record

Source and implementation. Anderson-APM is a transcription of `nearcorr_aa.m` from the code released with Higham and Strabić [18] (github.com/higham/anderson-accel-ncm). It keeps the order of the convergence test and the update, the bookkeeping of the stored history, the modified Gram–Schmidt update of the QR factors, MATLAB’s `qrdelete` for the oldest column, the optional column dropping (off by default) and the returned matrix. Only the default least-squares option, QR with updating, is transcribed; the paper’s experiments use it.

What was compared. Their Tables 1, 3, 4 and 7 report iteration counts on matrices that are public: four small matrices at history lengths $m = 1, \dots, 6$ and three eigenvalue floors δ (72 counts), two fixed-element problems at $m = 1, \dots, 5$ (10 counts), and `cor1399` and `cor3120` at $m = 2$. The RiskMetrics matrices and the $n = 90$ matrix of their Table 4 are not public. Their stopping test is $\|Y - X\|_F / \|Y\|_F < n \text{ eps}$.

Table 19: Published Anderson-acceleration iteration counts reproduced exactly, by LAPACK symmetric eigensolver. `syevr` is Julia’s default and is used for every result in the paper; `syevd` differs from it only in rounding.

counts	published	exact, <code>syevr</code>	exact, <code>syevd</code>
small matrices, excluding <code>mmb13</code>	64	57	61
<code>mmb13</code>	18	4	5
<code>cor1399</code> , <code>cor3120</code> ($m = 2$)	2	0	2
unaccelerated, <code>cor1399</code> , <code>cor3120</code>	2	2	2

Why counts differ. Under the default eigensolver the two large counts are 118 and 163, against published values of 124 and 174. Two checks separate an error of transcription from rounding at the stopping tolerance. First, an independent implementation that solves each least-squares problem from scratch, without updating, gives the same iterates as ours to within 2.3×10^{-12} in relative Frobenius norm until the stopping quantity reaches 10^{-10} , and the same iteration count at the tolerances 10^{-8} and 10^{-10} in all 82 small cases. Second, changing only the eigensolver moves the counts at $n \text{ eps}$ onto the published ones: under `syevd` both large counts are reproduced exactly, and 61 of the 64 small ones; the three that remain are one iteration apart. On `cor1399` the stopping quantity spends its last five iterations between one and four times the tolerance, so the step at which it first falls below is decided by the last bits of each eigendecomposition.

The matrix `mmb13`. This is the order-6 matrix on which unaccelerated alternating projections needs 801 iterations. Near convergence the stopping quantity stays within a factor of three of the tolerance for tens to hundreds of iterations, so the count at which it first falls below is decided by rounding, and the gaps are large, up to 56 iterations: at $\delta = 0$ and $m = 4$ the published count is 126 and ours 70. The two independent implementations disagree with each other by similar amounts

on this matrix while agreeing on its iterates to 10^{-12} , and the published counts are themselves not monotone in m (117 at $m = 3$, 126 at $m = 4$). We regard the counts on this matrix as uninformative about correctness either way.

What is claimed. We claim agreement of the iteration, not of every published count: our iterates agree with an independent implementation, and our counts agree with the published ones wherever rounding at the stopping tolerance does not decide them. No result in this paper depends on those counts. The ranking study stops at $\|\text{diag}(X) - e\|_2 \leq 10^{-11}$ and scores the recorded trajectory, and it uses the default eigensolver, as every other solver does. The comparison scripts and their logs are listed in section D.

D Artifact manifest

The generator, the solvers, the analysis scripts and every result file behind every table are available at <https://github.com/Realosunboy6/ncm-exact-benchmark>. Paths below are relative to the artifact root. An archived snapshot carrying a permanent identifier will be deposited on acceptance.

Instance binaries are not stored in full, since they reach 1 GB at $n = 500$; the manifests record a hash and four numerical invariants of every instance used. The generator is seeded and reproduces those instances up to rounding: regenerated on the original machine, each G differs from the original by at most 10^{-17} in relative Frobenius norm and each X^* is bit-identical, although the hashes, taken on exact bits, differ. The script `code/check_instances.py` compares a regenerated set with the recorded invariants instead. On another LAPACK build the generator may receive a basis for the degenerate kernel of X^* in a rotated form, which yields different members of the same family; the check detects this, since a rotated basis moves the invariants by 15 to 20 per cent. A 12-instance subset of the original instances at $n = 100$ (2 MB) ships with the artifact.

file	contents
<i>Instances</i>	
code/gen_degeneracy_family.py	generator and screening (section 2)
code/materialize_exact_ystars.py	exact $y^* = -\text{diag}(W)$ for a suite
data/canonical_n100/	12 instances with G , X^* , y^* , manifest and cases
data/manifests/	SHA-256 manifests, numerical fingerprints and screening logs for every set
code/check_instances.py	checks a regenerated set against the fingerprints
code/export_real_suite.py	literature matrices from Anymatrix CORRINV
code/export_thesis_kkt.py,	equity-derived sets (sections 5.8 and 5.9)
export_thesis_suite.py	
<i>Solvers</i>	
code/bench_sbb_dual.jl	sbb_dual, newton_ncm_globalized, agd_sdaj_ncm, dykstra_apm, trajectory recorder, study drivers
code/anderson_apm.jl	Anderson-APM, transcribed from Higham and Strabić's released code
code/agd_sdaj.jl	standalone AGD-SDAJ used for verification (section B)
code/validation/	checks against Higham (2002), Higham–Strabić (2016), Borsdorf (2007), Huynh–Hwang (2025), including the Anderson-APM record of section C; logs in results/validation/
code/highrank_queue.sh,	drivers for the runs added with Anderson-APM
anderson_rest_queue.sh	
code/check_reference_accuracy.jl	two independent references per matrix (section 5.10)
<i>Results and analysis</i>	
code/reproduce.sh	regenerates the analysis tables from results/*.csv.gz and the figure data, and prints which file holds which table
results/ranking_kkt270_v2.csv.gz	one row per EVD, the five original solvers, 270 instances at $n = 100$
results/ranking_n500.csv.gz	the same at $n = 500$
results/ranking_highrank_n500.csv.gz	$r \in \{75, 100, 150\}$, Newton-SIN-BH, AGD-SDAJ-BH and SBB-Dual (table 8)
results/ranking_highrank_proj.csv.gz	the same instances, Dykstra-APM and Anderson-APM
results/ranking_n500_dykstra1200.csv.gz	the 90 censored Dykstra-APM runs at $n = 500$ rerun at a cap of 1200 (section 5.11)
results/ranking_*anderson*.csv.gz	Anderson-APM on every other test set, run after the original studies
results/ranking_thesis*.csv.gz	equity sets at $n = 550$ (tables 9 and 10)
results/ranking_us2105_*.csv.gz	equity sets at $n = 2105$ (tables 11 and 12)
results/ranking_real.csv.gz	four literature matrices (table 13)
results/timing_*.csv	single-threaded timings of all seven variants, measured in one session (section 3.5)
results/timing_session1/	an earlier timing session, kept for comparison; not used in any table
code/analyze_ranking.py	tables 5 to 7, 10 and 14 and the accounting numbers of section 5.11
code/analyze_by_rank.py	tables 8, 9 and 11
code/analyze_per_instance.py	table 13
code/analyze_stopping_rule.py	the cross-dimension comparison of section 6
code/make_figure_data.py	the data behind every figure
code/analyze_highrank.py,	bootstrap intervals for table 8, table 12, and the
analyze_us2105.py, analyze_confound.py	bulk-level pilot
code/check_scoring_column.py	the rescaled-iterate sensitivity check (section 5.11)
code/analyze_timing.py	tables 2, 3 and 15

Data not redistributed. The Anymatrix CORRINV matrices carry their own terms and are obtained from their own repository. The two equity panels are built from vendor price data that may not be redistributed; the exporters, the selection rules and all result files are included, the raw prices are not.

Figure data. The files `paper/figures/data/*.dat` behind every figure are written by `code/make_figure_data.py` from the result files above, and `reproduce.sh` regenerates them for comparison. The file `evd_vs_eps.dat` applies the cost-to-target rule of `analyze_ranking.py` (accepted-only, reach and hold) on a half-decade grid of ε ; at the five decade targets it reproduces table 5, up to that table’s rounding of half-integer medians. The data behind figs. 1 and 5, like tables 2, 3 and 15, come from the timing runs.

E Solver pseudocode

The four solvers of section 5.1 as implemented. Parameter values and the sources they are transcribed from are given there; the verification record for AGD-SDAJ is section B.

Algorithm 1 Newton-SIN-BH (semismooth Newton, cached factorization, BH line search)

Require: G , tolerance τ ; $\sigma = 10^{-4}$, $\mu_{\text{BH}} = 0.9$, $\gamma = 100$, $u = 2^{-53}$

```

1:  $y \leftarrow 0$ ;  $(\theta, g, X, \Lambda, P) \leftarrow \text{EVAL}(y)$  ▷ keep the factorization
2: while  $\|g\| > \tau$  do
3:   Solve  $(V + 10^{-12}I)d = -g$  by CG, with  $V$  the generalized Jacobian from the cached  $(\Lambda, P)$ ;
   stop at  $\|r\| \leq \min\{0.1, \|g\|\}\|g\|$  or on negative curvature
4:   if  $d$  not finite or  $g^\top d \geq 0$  then  $d \leftarrow -g$ 
5:   end if
6:    $\alpha \leftarrow 1$ 
7:   loop
8:      $(\theta_t, g_t, X_t, \Lambda_t, P_t) \leftarrow \text{EVAL}(y + \alpha d)$ 
9:     if  $\theta_t \leq \theta + \sigma \alpha g^\top d$  then accept; break
10:    end if
11:    if  $|\theta_t - \theta| < \gamma u(1 + |\theta_t| + |\theta|)$  then ▷ BH near-equality branch
12:      if full step has  $\|g_t^{(\alpha=1)}\| \leq (1 - \mu)\|g\|$  then accept the full step; break
13:      end if
14:      accept the unit steepest-descent step  $y - g$  (one further EVD); break
15:    end if
16:     $\alpha \leftarrow \alpha/2$ 
17:  end loop
18:  update  $(y, \theta, g, X, \Lambda, P)$  from the accepted trial ▷ no recomputation
19: end while
20: return  $X$ 

```

Algorithm 2 AGD-SDAJ [19] (AGD-SDAJ-BH: with the marked branch)

Require: $G, \tau; q = 2, c = 10^{-4}, \rho = 0.5, \lambda_1 = \lambda_2 = 0.01, \epsilon_l = 10^{-5}, \epsilon_u = 10^8$

```
1:  $y \leftarrow 0; (\theta, g, X) \leftarrow \text{EVAL}(y)$ 
2: while  $\|g\| > \tau$  do
3:    $(y^p, \theta^p, g^p) \leftarrow (y, \theta, g); b \leftarrow e$  ▷  $B_0 = I$ , reset every outer iteration
4:   for  $i = 0, \dots, q - 1$  while  $\|g^p\| > \tau$  do ▷ QN-SDAJ inner steps
5:      $d \leftarrow -g^p \oslash b; \epsilon_i \leftarrow 1/(i+1)^2; \alpha \leftarrow 1$ 
6:     backtrack  $\alpha \leftarrow \rho\alpha$  until, with  $g_t = \nabla\theta(y^p + \alpha d)$ ,  

 $\|g_t\|^2 \leq (1 + \epsilon_i)\|g^p\|^2 - \lambda_1\alpha^2\|g^p\|^2 - \lambda_2\alpha^2\|d\|^2$ 
    ▷ eq. (6)
7:      $\beta \leftarrow a/b'$  with  $a = -g^{p\top}d, b' = (g_t - g^p)^\top d$  if  $b' > 0$ , else  $\beta \leftarrow 1$  ▷ general direction
8:      $s \leftarrow \beta\alpha d; y^+ \leftarrow y^p + s$ ; evaluate at  $y^+$  (reuse the trial if  $\beta = 1$ )
9:      $b_j \leftarrow z_j/s_j$  if  $z_j/s_j \in [\epsilon_l, \epsilon_u]$ , else 1, where  $z = g^+ - g^p$ 
10:     $(y^p, \theta^p, g^p) \leftarrow (y^+, \theta^+, g^+)$ 
11:  end for
12:  if  $\theta^p > \theta$  then  $(y^p, \theta^p, g^p) \leftarrow (y, \theta, g)$ 
13:  end if
14:  if  $\|g^p\| \leq \tau$  then  $y \leftarrow y^p$ ; break ▷ early exit at  $x^{\text{pre}}$ 
15:  end if
16:   $d \leftarrow -g^p; \alpha \leftarrow 1$ 
17:  loop ▷ Algorithm 1, line 5
18:     $(\theta_t, g_t) \leftarrow \text{EVAL}(y^p + \alpha d)$ 
19:    if  $\theta_t \leq \theta^p + c\alpha g^{p\top}d$  then break
20:    end if
21:    if BH and  $|\theta_t - \theta^p| < \gamma u(1 + |\theta_t| + |\theta^p|)$  and  $\|g_t\| < \|g^p\|$  then break
22:    end if
23:     $\alpha \leftarrow \rho\alpha$ 
24:  end loop
25:   $\beta \leftarrow \|g^p\|^2 / ((g^p - g_t)^\top g^p)$  if the denominator is  $> 0$ , else 1
26:   $y \leftarrow y^p - \beta\alpha g^p; (\theta, g, X) \leftarrow \text{EVAL}(y)$  (reuse the trial if  $\beta = 1$ )
27: end while
28: return  $X$ 
```

Algorithm 3 SBB-Dual (clamped spectral gradient on the dual)

Require: $G, \tau, \eta = 0.01$

```
1:  $y \leftarrow 0; (\theta, g, X) \leftarrow \text{EVAL}(y)$ 
2: while  $\|g\| > \tau$  do
3:   if first update, or  $s^\top w \leq 0$ , where  $s = y - y_{\text{prev}}, w = g - g_{\text{prev}}$  then
4:      $t \leftarrow 1$ 
5:   else
6:      $t \leftarrow \min\{\max\{s^\top s / s^\top w, \eta\}, 2 - \eta\}$  ▷ clamped BB1
7:   end if
8:    $(y_{\text{prev}}, g_{\text{prev}}) \leftarrow (y, g); y \leftarrow y - tg; (\theta, g, X) \leftarrow \text{EVAL}(y)$ 
9: end while
10:  $X \leftarrow D^{-1/2} X D^{-1/2}, D = \text{Diag}(\text{diag } X)$ ; set  $\text{diag}(X) = e$  ▷ rescaling epilogue
11: return  $X$ 
```

Algorithm 4 Dykstra-APM (alternating projections with Dykstra correction)

Require: G, τ

```
1:  $X \leftarrow G; \Delta S \leftarrow 0; \Delta U \leftarrow 0$ 
2: loop
3:    $Y \leftarrow \Pi_{\mathcal{S}_+^n}(X + \Delta S)$  ▷ one EVD
4:   if  $\|\text{diag}(Y) - e\|_2 \leq \tau$  then return  $Y$  ▷ PSD half-iterate
5:   end if
6:    $\Delta S \leftarrow X + \Delta S - Y$ 
7:    $Z \leftarrow Y + \Delta U$  with  $\text{diag}(Z) \leftarrow e; \Delta U \leftarrow Y + \Delta U - Z; X \leftarrow Z$ 
8: end loop
```

Algorithm 5 Anderson-APM (alternating projections with Anderson acceleration, history m ; Higham and Strabić [18])

Require: G, τ , history length m (here $m = 2$)

```
1:  $Y \leftarrow G; \Delta S \leftarrow 0; z \leftarrow (\text{vec } Y, \text{vec } \Delta S)$ 
2: for  $k = 1, 2, \dots$  do
3:    $X \leftarrow \Pi_{\mathcal{S}_+^n}(Y - \Delta S)$  ▷ one EVD
4:    $\Delta S' \leftarrow X - (Y - \Delta S); Y' \leftarrow X$  with  $\text{diag}(Y') \leftarrow e$ 
5:    $g \leftarrow (\text{vec } Y', \text{vec } \Delta S'); f \leftarrow g - z$ 
6:   if  $\|Y' - X\|_F / \|Y'\|_F \leq \tau$  then return  $X$  ▷ PSD half-iterate
7:   end if
8:    $m_k \leftarrow \min(m, k - 1)$ ; form  $\Delta F_k, \Delta G_k$  from the last  $m_k$  differences of  $f, g$ 
9:    $\gamma \leftarrow \arg \min_{\gamma} \|f - \Delta F_k \gamma\|_2$  ▷ QR with updating
10:   $z \leftarrow g - \Delta G_k \gamma$ ; unpack  $z$  into  $Y, \Delta S$ 
11: end for
```

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